PHYSIOLOGY OF THE MOTOR SYSTEM

Control of gaze

Lecture 03

A · The eyes

	Somatic	Eye
Effector	~200 joints, ~600 muscles	one sphere, three muscle pairs
Degrees of freedom	more than any task needs	yoked: direction + depth
Load	varies with what you hold	the orbit's own, predictable
Motor unit	up to ~1,800 fibres	~5 fibres
Feedback	muscle sense, touch, vision	vision; a different error for each

Kandel 6e, Chs. 30, 31 and 35 · teaching contrast, row by row; not a wiring diagram

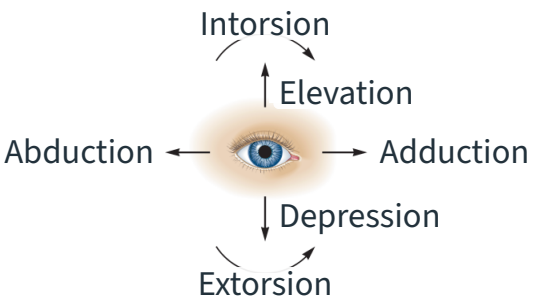
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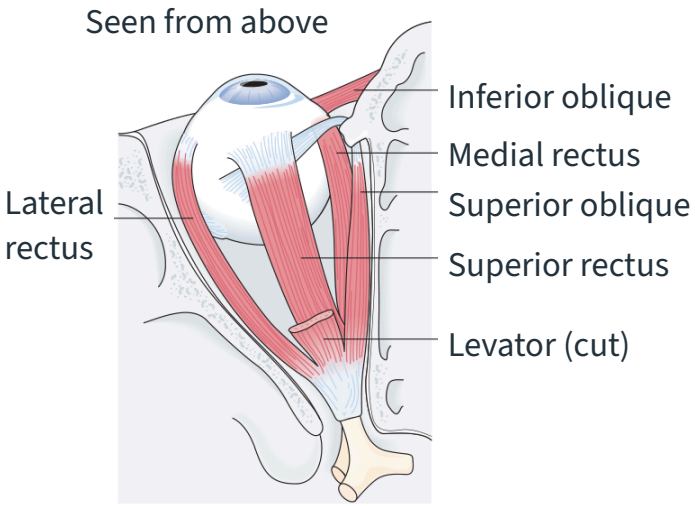
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> Control of eye movements must be different from control of arm movements.

The eye is controlled by 3 pairs of muscles

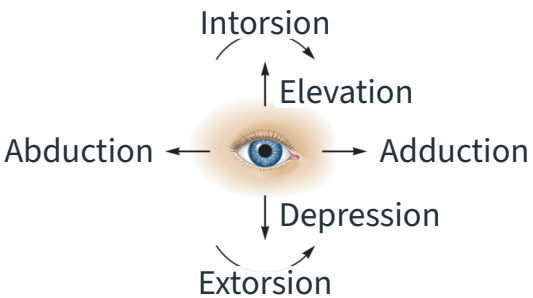


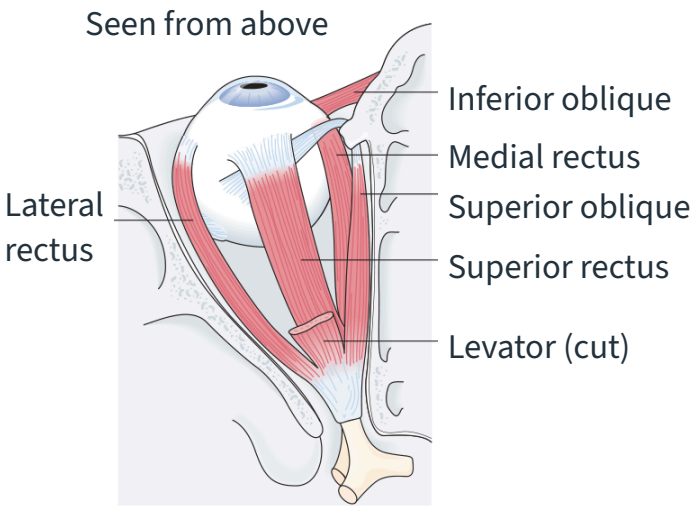


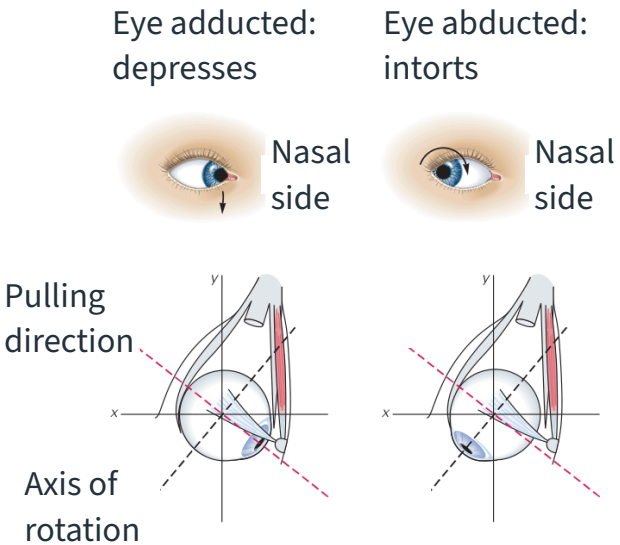
Kandel 6e, Figs. 35–1A and B2, p. 861 and 35–2, p. 862 · extraocular muscles; labels re-set from the figures

- Six muscles form three agonist–antagonist pairs, one for each rotation axis.

The eye is controlled by 3 pairs of muscles







Kandel 6e, Figs. 35–1A and B2, p. 861 and 35–2, p. 862 · extraocular muscles; labels re-set from the figures

- Six muscles form three agonist–antagonist pairs, one for each rotation axis.
- The superior oblique:
  - Depresses adducted eye
  - Intorts abducted eye
- So what a vertical muscle does depends on where the eye already points.

Wired yoking makes the two eyes move as one

Conjugate gaze to the right

Right eye	Left eye
lateral rectus	medial rectus
superior rectus (up)	inferior oblique (up)
inferior rectus (down)	superior oblique (down)

- Each muscle has a yoke partner in the other eye, and the two are driven together.

Kandel 6e, Ch. 35, pp. 862–863; Ch. 30, p. 714 · yoke pairs from the text; Table 35–1 gives the vertical muscles' actions

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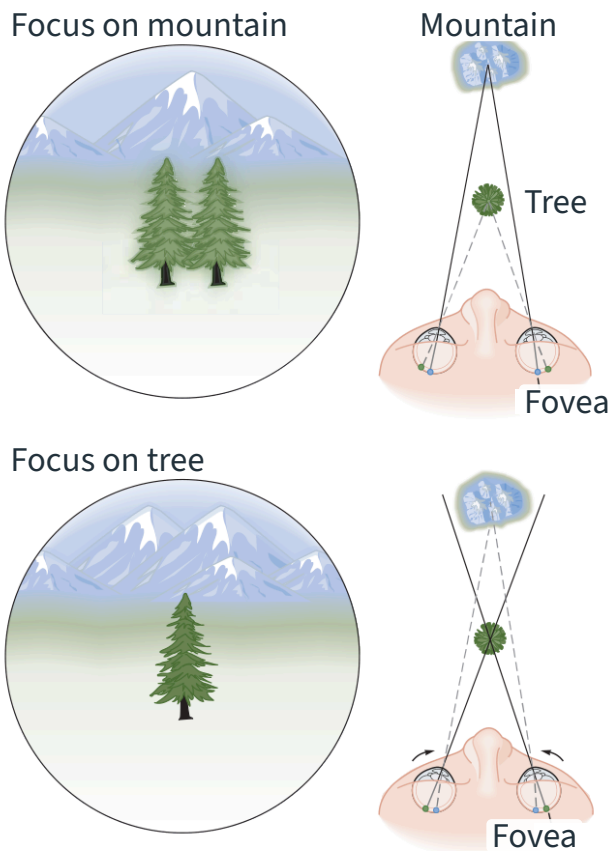
Degrees of freedom

- Yoking leaves **direction** and **depth** as the 3 main degrees of freedom
  - Direction: horizontal and vertical
  - Depth: vergence

- Each muscle has a yoke partner in the other eye, and the two are driven together.
- Extra degrees of freedom:
  - Eyes: removed by wiring, not control
  - Arm: more than any task needs; the controller must choose

Kandel 6e, Ch. 35, pp. 862–863; Ch. 30, p. 714 · yoke pairs from the text; Table 35–1 gives the vertical muscles' actions

Vergence movements break the yoke

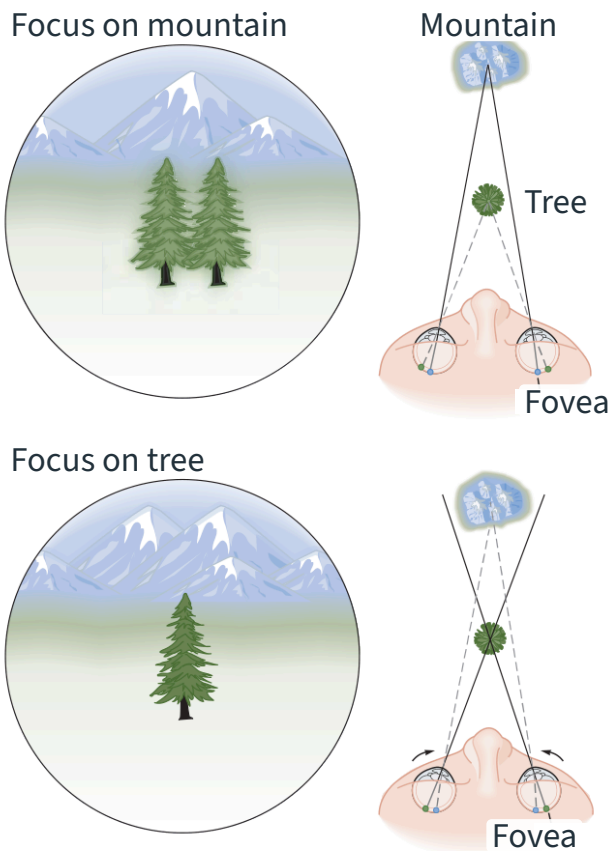


- Only the horizontal recti work

> - Far to near, the eyes turn in opposite directions.

Kandel 6e, Fig. 35–16, p. 879 · schematic after F. A. Miles; labels re-set from the figure

Vergence movements break the yoke



- Only the horizontal recti work

> - Far to near, the eyes turn in opposite directions.
> - It supplies depth, the freedom the yoke leaves.

Kandel 6e, Fig. 35–16, p. 879 · schematic after F. A. Miles; labels re-set from the figure

Eye muscles have tiny motor units

Innervation number

≈ 5
fibres per motor neuron, eye muscle

≈ 1,800
fibres per motor neuron, leg muscle

- Eye force is finely graded; fibres per motor neuron:
  - Eye: about 5
  - Leg: about 1,800

Kandel 6e, Table 31–1, p. 739; Ch. 30, p. 717

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The load

- Only the orbit's own viscosity and elasticity
  - No object to lift, no ground to push against
- The eye "tends not to be substantially perturbed by external events"

- Eye force is finely graded; fibres per motor neuron:
  - Eye: about 5
  - Leg: about 1,800
- With a predictable load, the force for a movement can be known in advance.
- So commands can be issued without waiting for feedback.

Kandel 6e, Table 31–1, p. 739; Ch. 30, p. 717

Section A summary: the eye is a simple, predictable effector

	Somatic	Eye
Effector	~200 joints, ~600 muscles	one sphere, three muscle pairs
Degrees of freedom	more than any task needs	yoked: direction + depth
Load	varies with what you hold	the orbit's own, predictable
Motor unit	up to ~1,800 fibres	~5 fibres
Feedback	muscle sense, touch, vision	vision; a different error for each

TAKE HOME

- Few degrees of freedom: yoked; vergence breaks the yoke.
- Tiny motor units and a predictable load: forces can be planned in advance.

Kandel 6e, Chs. 30, 31 and 35 · the section A table, returning

B · Gaze and reaching

	Somatic	Eye
Final common path	motor neurons and muscles	oculomotor nuclei and muscles
Systems	posture, locomotion, reach, grasp	six: one shift, five holds
Overlap	shared spinal interneurons	shared brainstem nuclei, FEF
Control	feedforward and feedback	feedforward and feedback
Response latency	25 ms (reflex) to 200 ms (visual)	15 ms (VOR) to 200 ms (saccade)

Kandel 6e, Chs. 30, 33 and 35; Kandel 5th ed., Ch. 40; response latencies: instructor, 2026-11-08 · teaching comparison, row by row; not a wiring diagram

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> Control of gaze is organized like control of the body.

One set of muscles serves many kinds of movement

Body

- Functions: eye movement, reach and grasp, posture, locomotion, breathing, speech
  - Served by overlapping groups of muscles
- Breathing muscles, in every control mode:
  - Voluntary: deep breath before a dive
  - Rhythmic: regular breathing
  - Reflexive: cough

> - Body: one set of muscles serves many functions and all three control modes.

Kandel 6e, Ch. 30, p. 715; Ch. 35, p. 866

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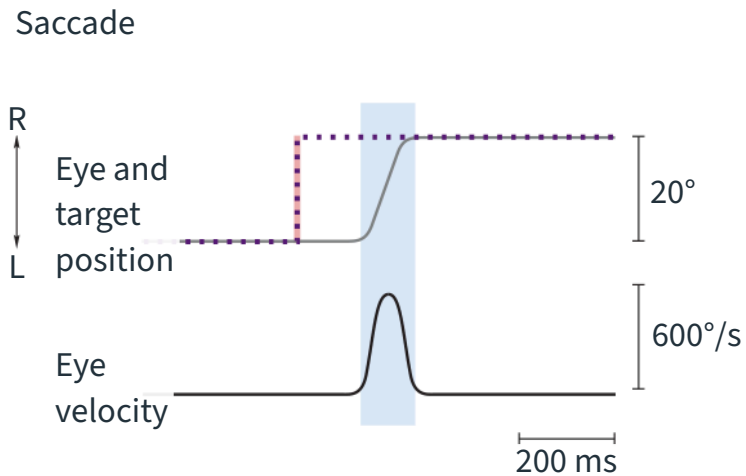
Eye

- Fixating, shifting, following, converging, stabilizing
- The oculomotor nuclei are the final common targets of every type of eye movement

- Body: one set of muscles serves many functions and all three control modes.
- Eye: every type of movement ends on the same motor neurons.
- In both, many systems converge on one final common path: the motor neurons and their muscles.

Kandel 6e, Ch. 30, p. 715; Ch. 35, p. 866

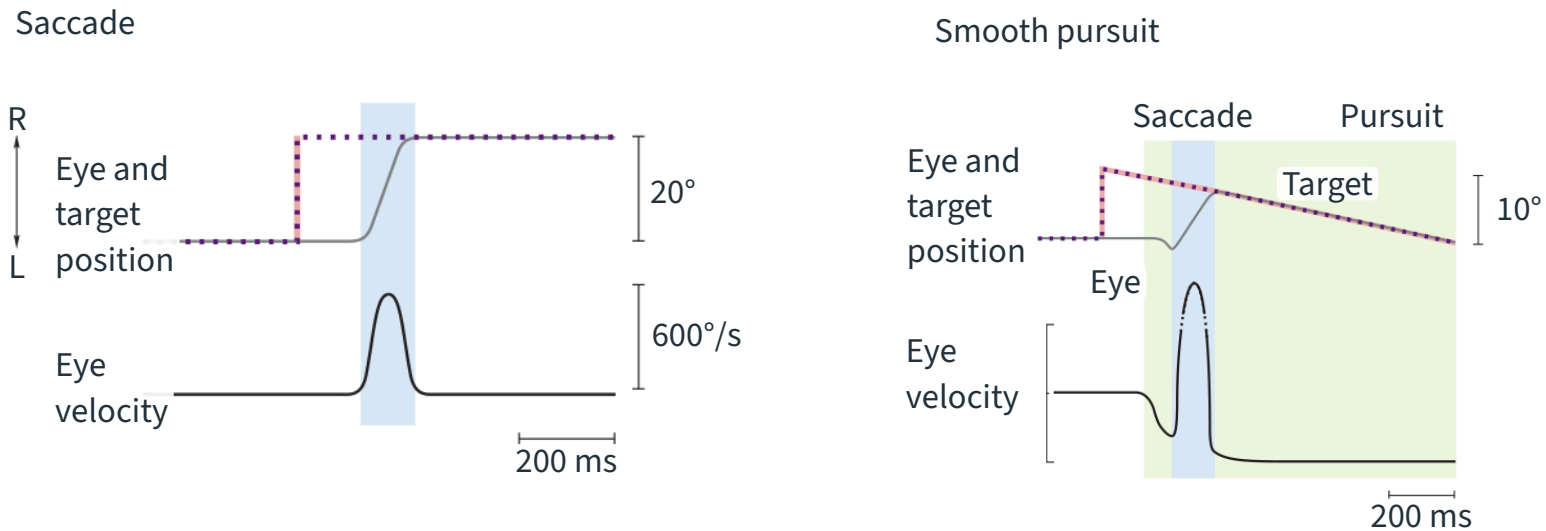
The saccade system and the smooth pursuit system work together



Kandel 6e, Fig. 35–6, p. 867 · human eye position and velocity; A a saccade, B smooth pursuit with a catch-up saccade; labels re-set from the figure

- The saccade:
  - Brief jump with one sharp velocity peak
  - Over before vision can correct it

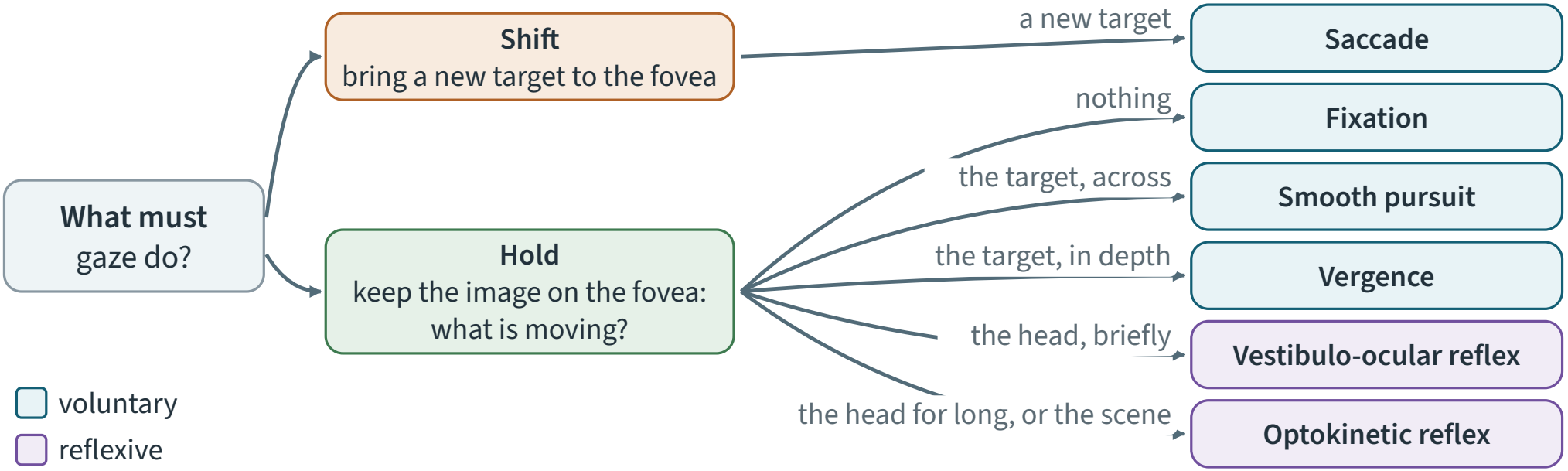
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- The saccade:
  - Brief jump with one sharp velocity peak
  - Over before vision can correct it
- Pursuit:
  - Eye velocity matches target velocity
  - Catch-up saccade removes the position error

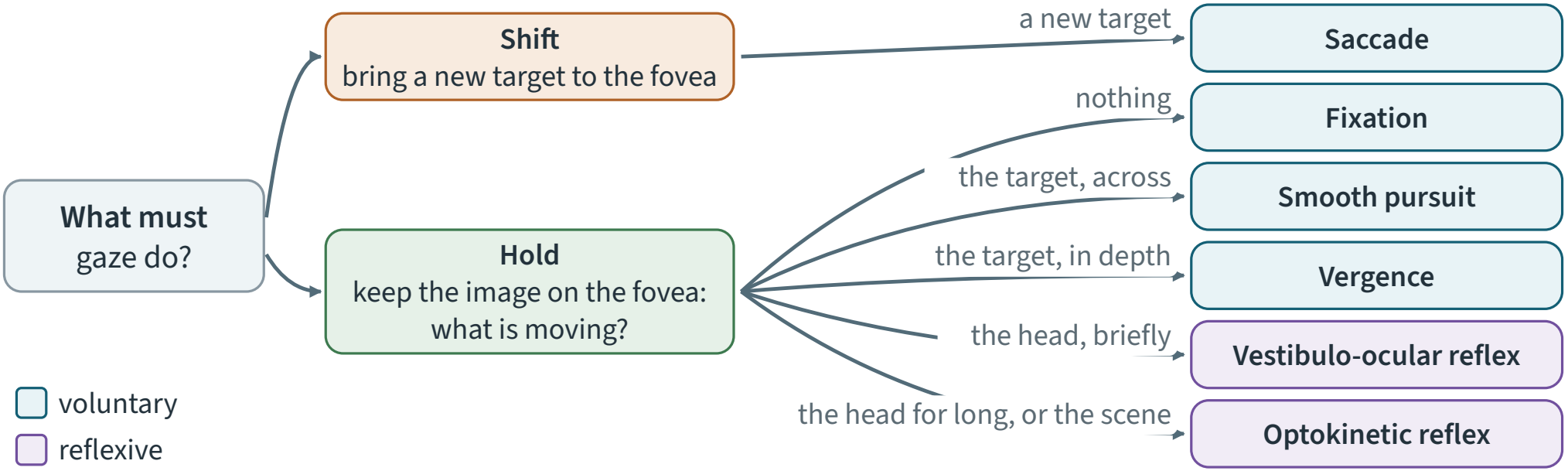
Gaze has six systems



Kandel 6e, Ch. 35, p. 866 · the six systems from the text; the shift/hold grouping and the control-mode colours are a teaching synthesis. Body: L01′s circuit maps, Kandel 6e, Figs. 33–1, 34–2A, 36–13

- Eye: six systems, one per disturbance.

Gaze has six systems



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- Eye: six systems, one per disturbance.
- Body, as in L01: posture, locomotion, reach and grasp each have their own circuits.

Body and eye systems partly overlap

Body

- Spinal interneurons of the locomotor rhythm:
  - Gate reflexes: the same stimulus excites flexors in one phase, extensors in another
  - Carry brainstem and cortical commands to the muscles
- Posture joins voluntary action: ankle torque changes before an arm pull

> - Body: reflex, rhythmic and voluntary control share spinal circuits.

Kandel 6e, Ch. 33, pp. 798–799, 803–805; Fig. 36–8, p. 893 (L01); Ch. 35, pp. 869–870, 873, 875, 878–879; Kandel 5th ed., Ch. 40, pp. 926–928

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Eye

- Medial vestibular nucleus and prepositus:
  - Integrator for saccades
  - Eye-velocity neurons for pursuit
  - First relay of the vestibulo-ocular reflex
- The same vestibular nucleus neurons respond to head rotation and to scene motion
- FEF: saccade cells and a pursuit region
- Collicular fixation zone excites omnipause neurons, which gate saccades

- Body: reflex, rhythmic and voluntary control share spinal circuits.
- Eye: the systems share brainstem nuclei and cortical areas.
- In both, the systems are partly distinct and partly shared.

Kandel 6e, Ch. 33, pp. 798–799, 803–805; Fig. 36–8, p. 893 (L01); Ch. 35, pp. 869–870, 873, 875, 878–879; Kandel 5th ed., Ch. 40, pp. 926–928

Body and eye combine feedforward and feedback

Feedforward: no error used during the movement

Saccade	visual error, afterwards
Vestibulo-ocular reflex	image slip, over days

Both feedforward errors are used later,
to recalibrate the command.

Feedback: a visual error, used throughout

Fixation	position error on the fovea
Smooth pursuit	retinal slip of the target
Vergence	disparity between the eyes
Optokinetic reflex	retinal slip of the scene

Kandel 6e, Ch. 30, pp. 717–719, 735; Ch. 35, pp. 866–867, 878–880; Kandel 5th ed., Ch. 40, pp. 924, 930 · teaching synthesis

- Eye: saccade and vestibulo-ocular reflex feedforward; the holding systems on visual feedback.

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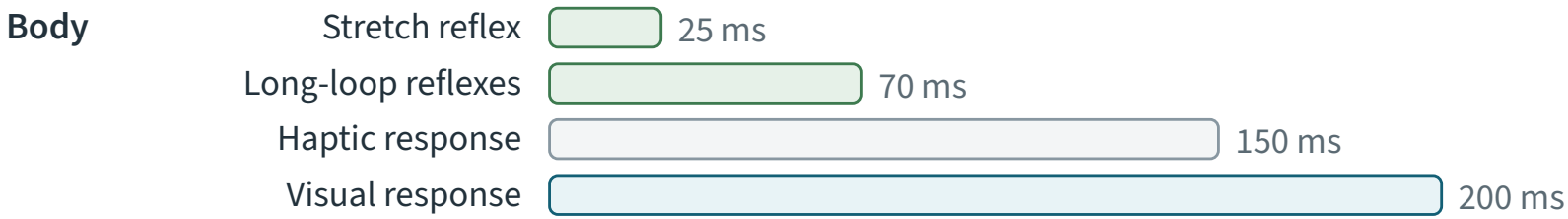
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- Eye: saccade and vestibulo-ocular reflex feedforward; the holding systems on visual feedback.
- Body, as in L01′s reach and grip: feedforward first, feedback after.

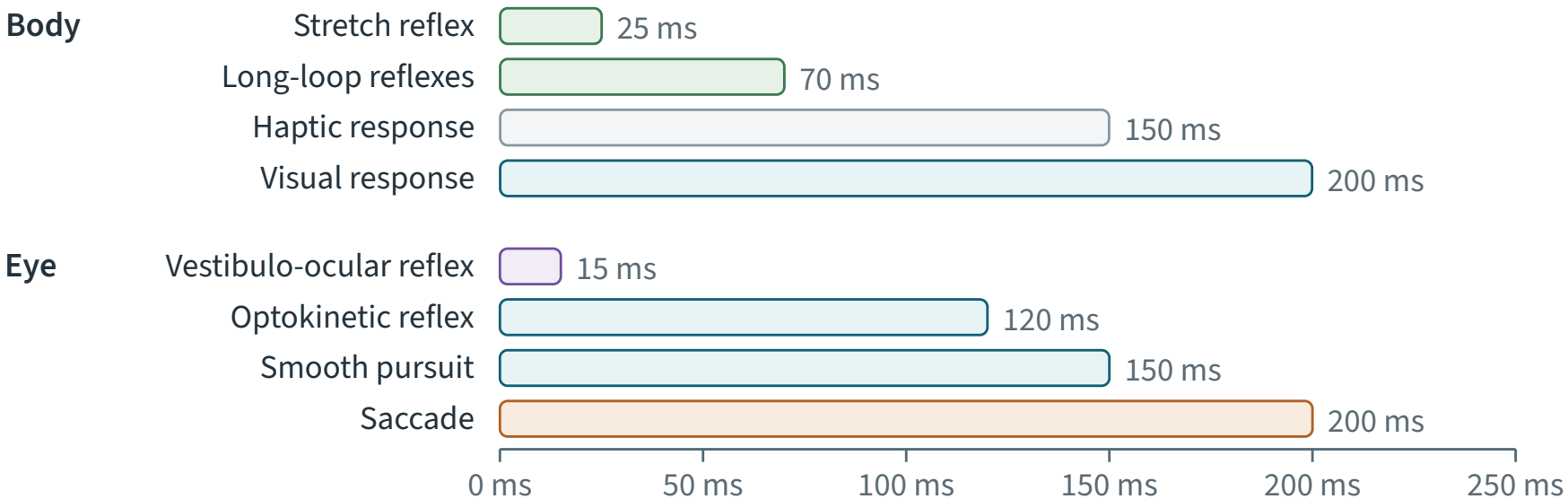
Multiple feedback loops control body and gaze



Response latencies: instructor, 2026-11-08. Saccade duration: Kandel 6e, Ch. 30, p. 717

- Body: reflexes in 25–70 ms; haptic and visual responses in 150–200 ms.

Multiple feedback loops control body and gaze



Response latencies: instructor, 2026-11-08. Saccade duration: Kandel 6e, Ch. 30, p. 717

- Body: reflexes in 25–70 ms; haptic and visual responses in 150–200 ms.
- Eye:
  - Vestibulo-ocular reflex: 15 ms, the fastest
  - Optokinetic reflex, pursuit, saccade: 120–200 ms, visual delays
  - A saccade lasts ≈30 ms, too brief to steer

Section B summary: gaze is organized like body movement

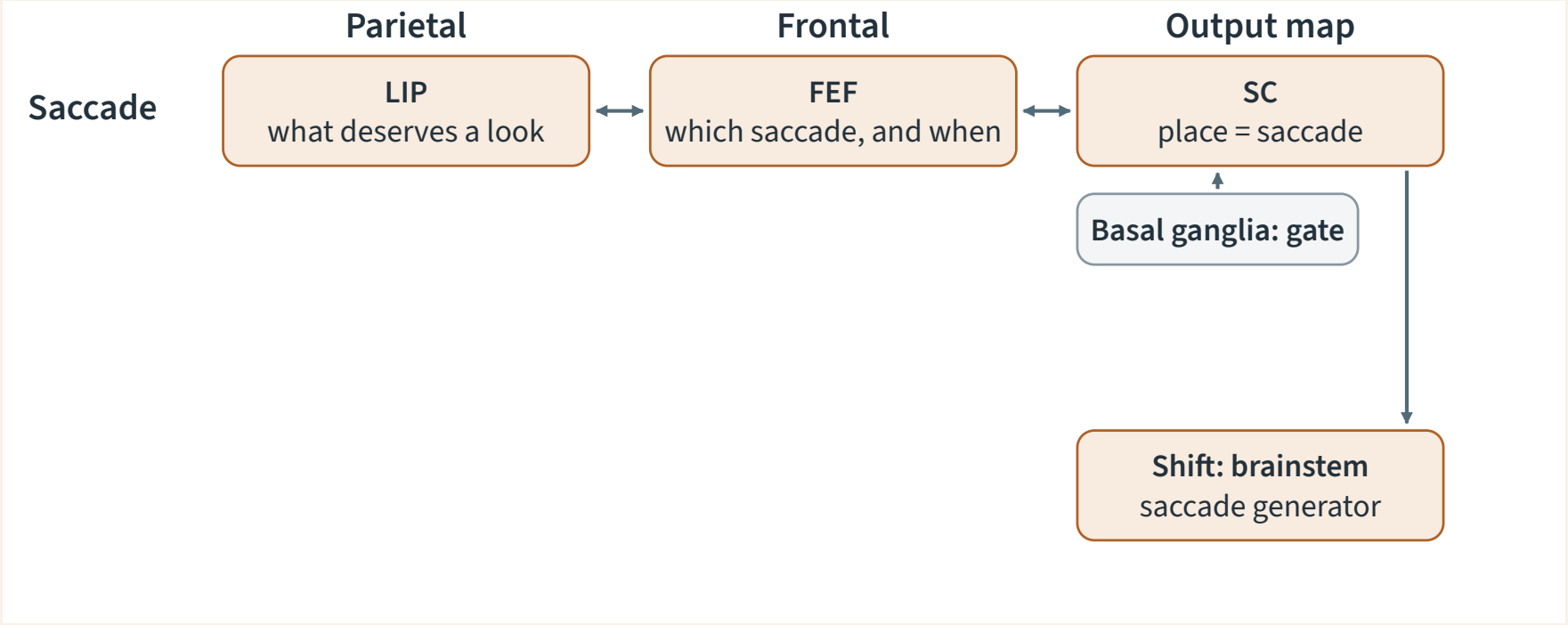
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TAKE HOME

- One final common path; different, partly overlapping systems.
- Feedforward and feedback, on loops with different latencies.

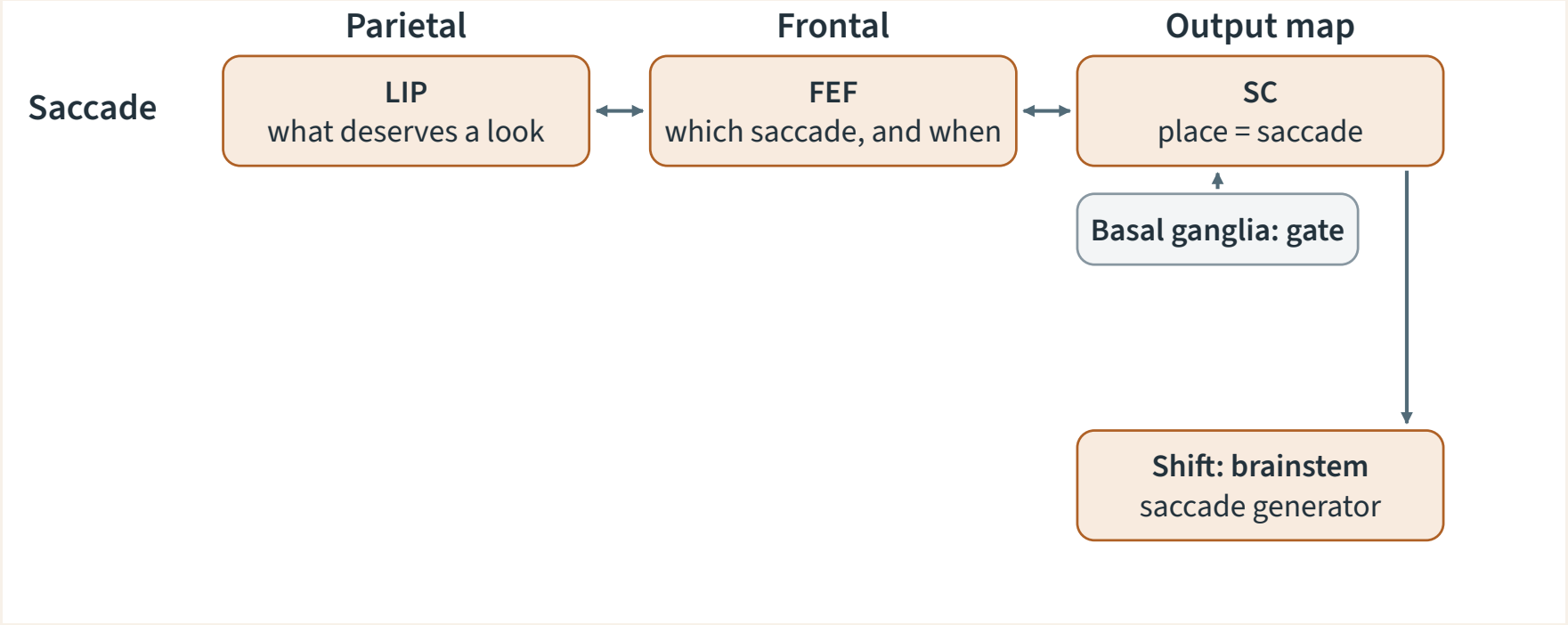
Kandel 6e, Chs. 30, 33 and 35; response latencies: instructor, 2026-11-08 · the section B table, returning

C · Saccade target selection



Kandel 6e, Ch. 35 · lecture organizer, built up section by section; a teaching schematic, not measured wiring

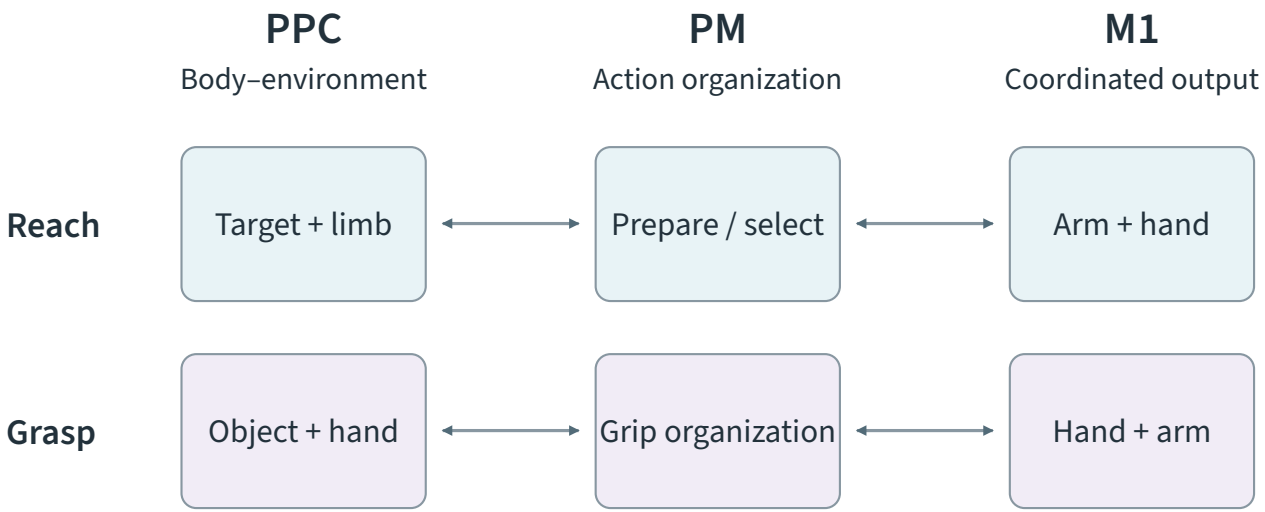
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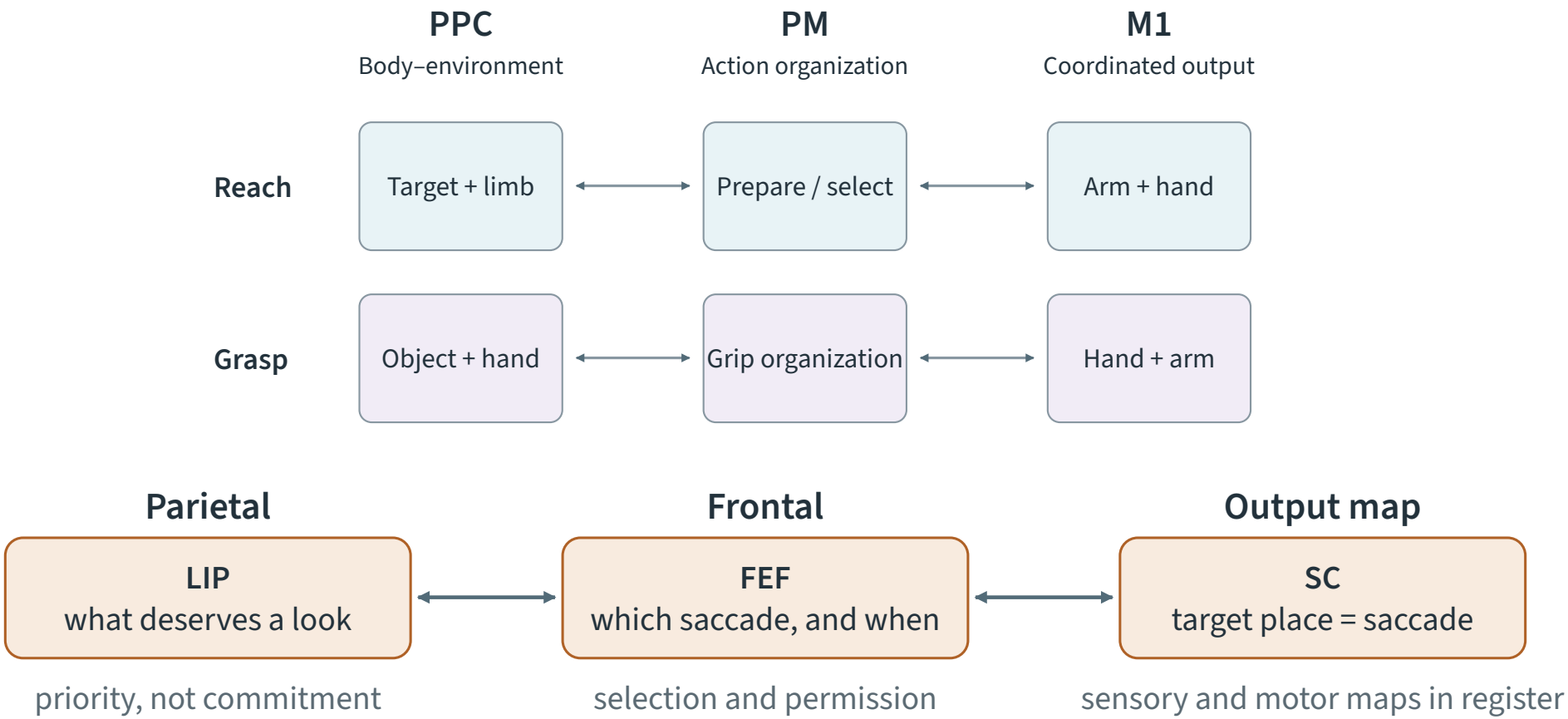
- Parietal and frontal cortex pick the target.
- The colliculus turns its place into a saccade, gated by the basal ganglia.

Saccades and reaches share three stages



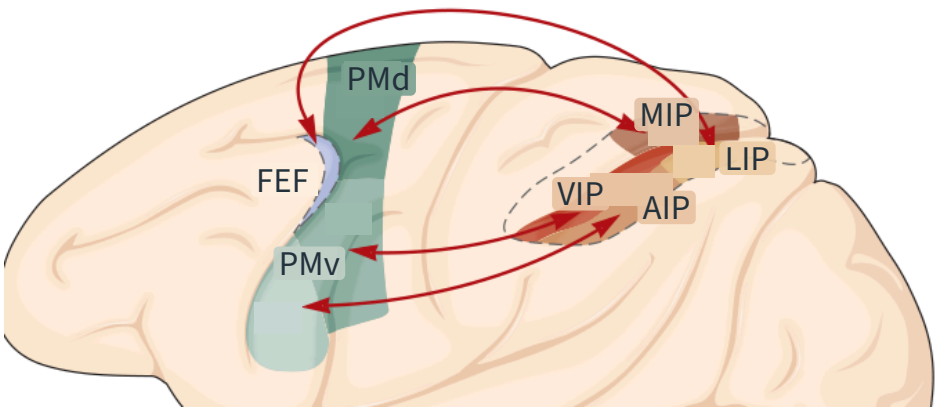
Kandel 6e, Chs. 34 and 35 · two teaching schematics in the same form: PPC / premotor / M1 above, LIP / FEF / SC below

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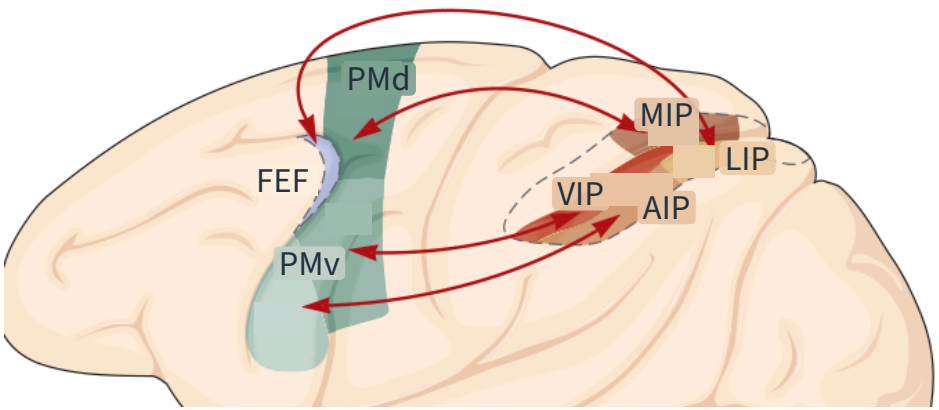
LIP and FEF form a parietal–frontal pair

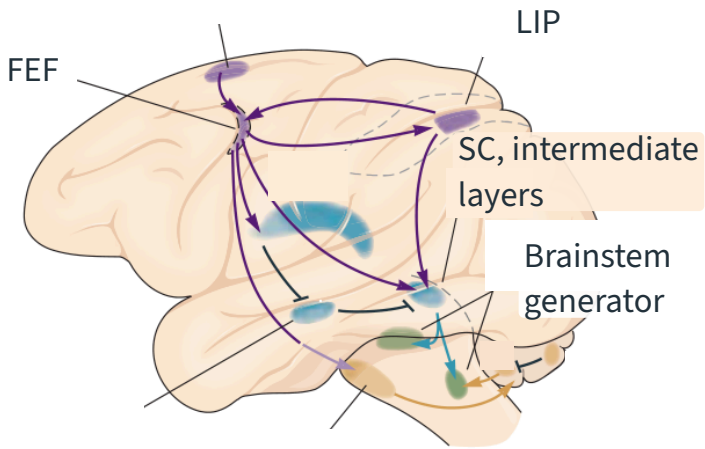


Kandel 5th ed., Fig. 29–14, p. 650 · intraparietal areas and their frontal partners; Kandel 6e, Fig. 35–9A, p. 871 · saccade pathways in the monkey; labels re-set from the figures

- Parietal → frontal pairs:
  - LIP → FEF: gaze
  - MIP → PMd: reach
  - AIP → PMv: grasp

LIP and FEF form a parietal–frontal pair

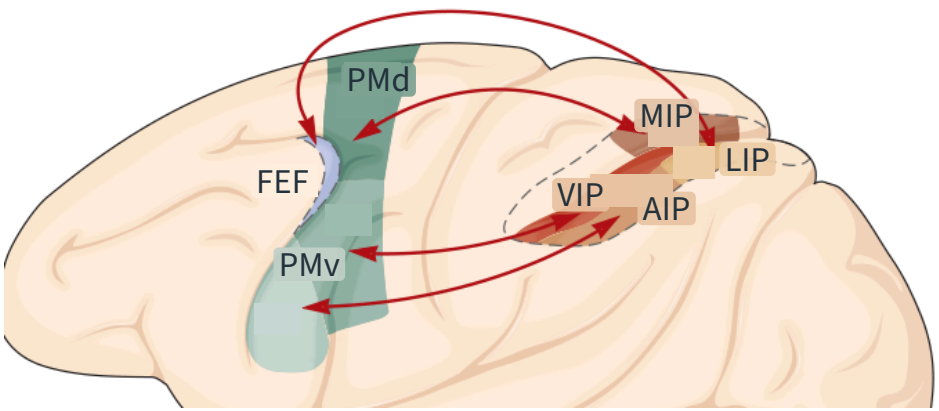


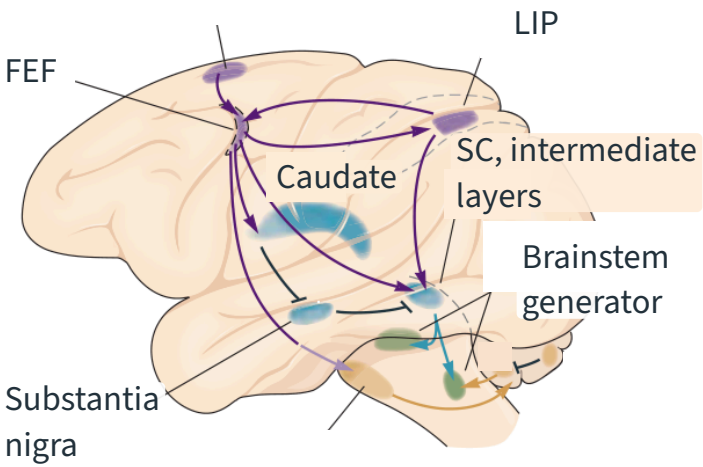


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- LIP and FEF both project to the colliculus, which drives the brainstem.

LIP and FEF form a parietal–frontal pair

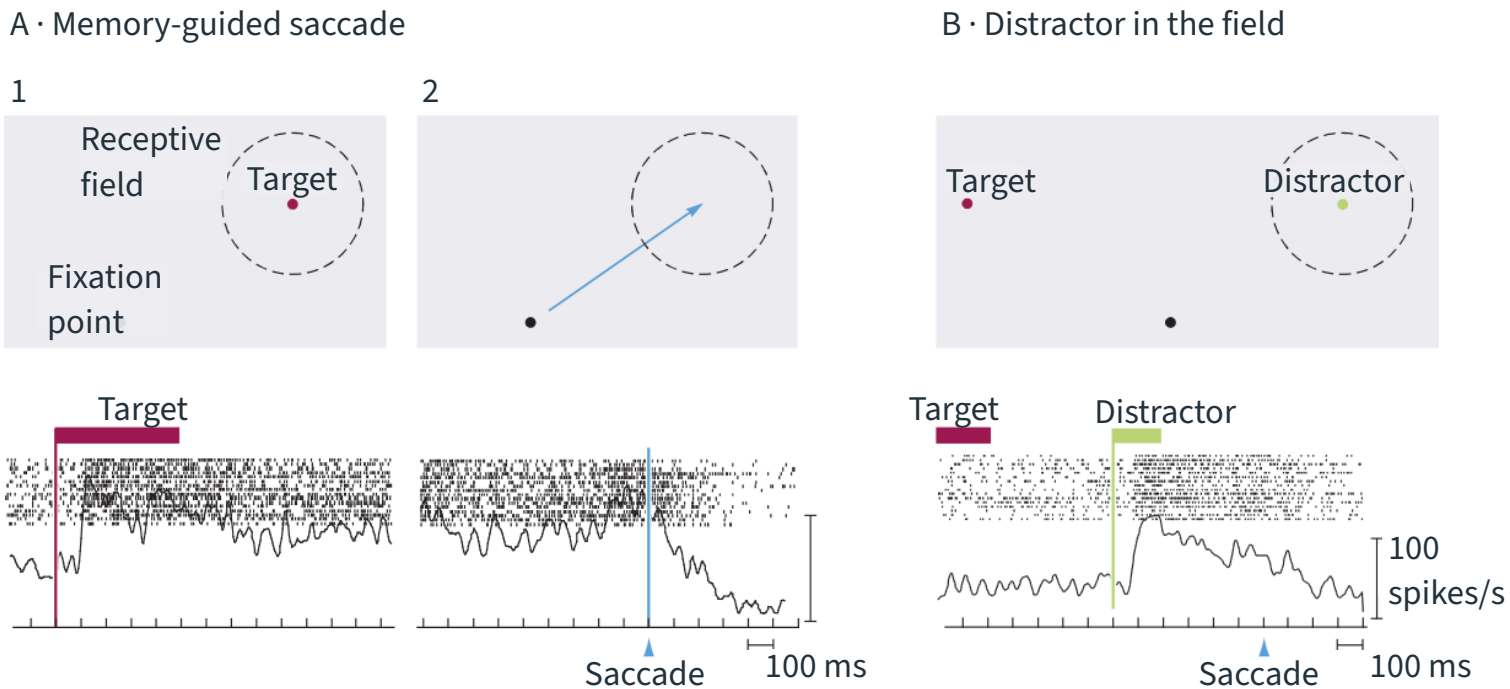




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- LIP and FEF both project to the colliculus, which drives the brainstem.
- The basal ganglia sit on the way: caudate, then substantia nigra.

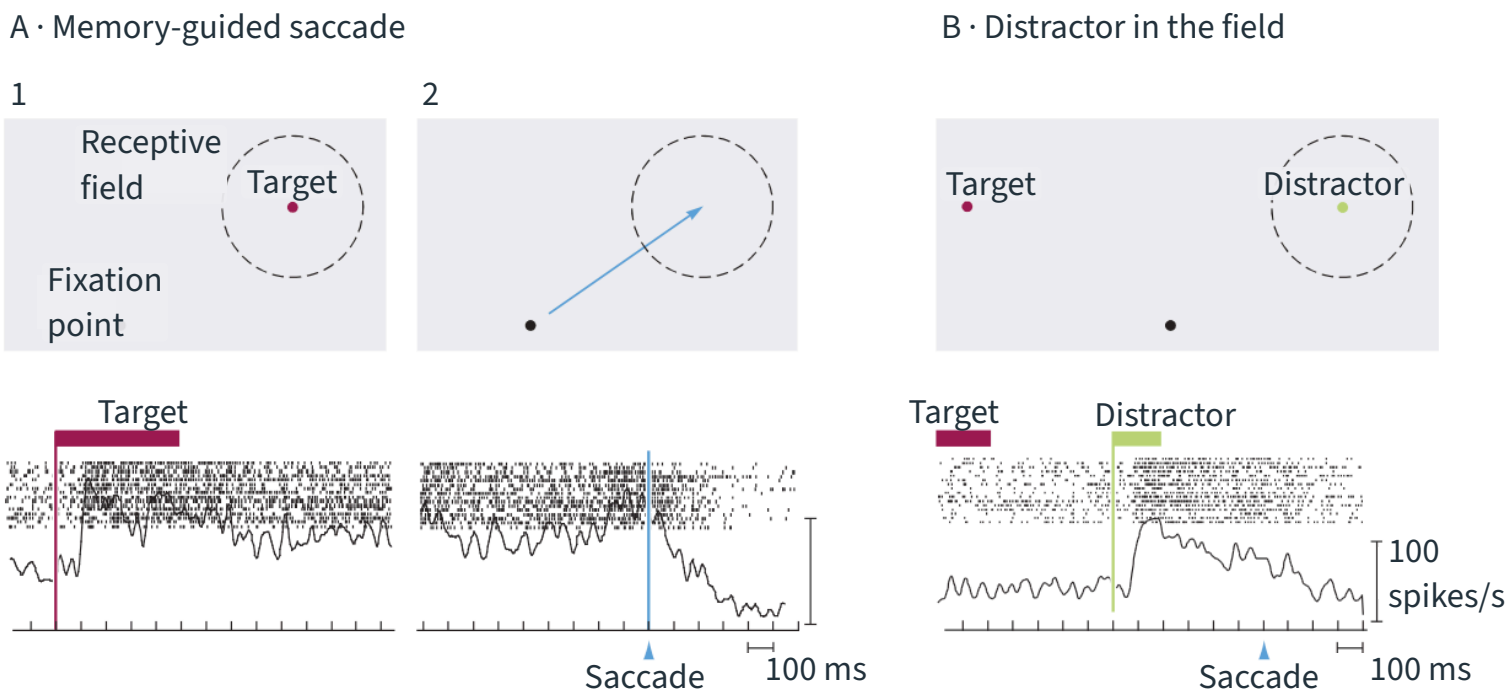
LIP neurons fire for targets and distractors



Kandel 6e, Fig. 35–12, p. 874 · single LIP neuron, memory-guided saccade task (Powell & Goldberg 2000); panel headings replaced, they state the result

- The cell fires from target to saccade, and as much for a distractor.

LIP neurons fire for targets and distractors



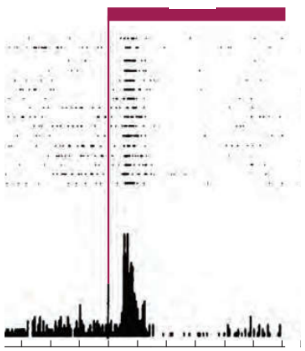
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- The cell fires from target to saccade, and as much for a distractor.
- So LIP marks **priority**, not a committed plan.

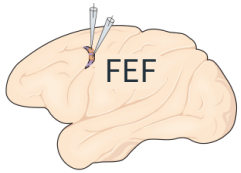
Aligning on two events shows what an FEF cell encodes

Target

Cell A



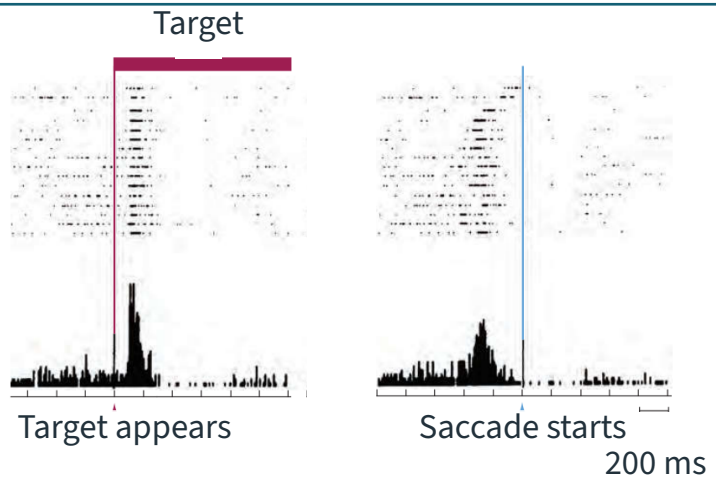
Target appears

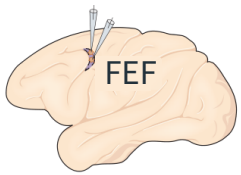


- Stimulating FEF evokes saccades

Kandel 6e, Fig. 35–13, p. 876 · two FEF neurons (Bruce & Goldberg 1985); panel headings removed, they state the answer; labels re-set from the figure

Aligning on two events shows what an FEF cell encodes

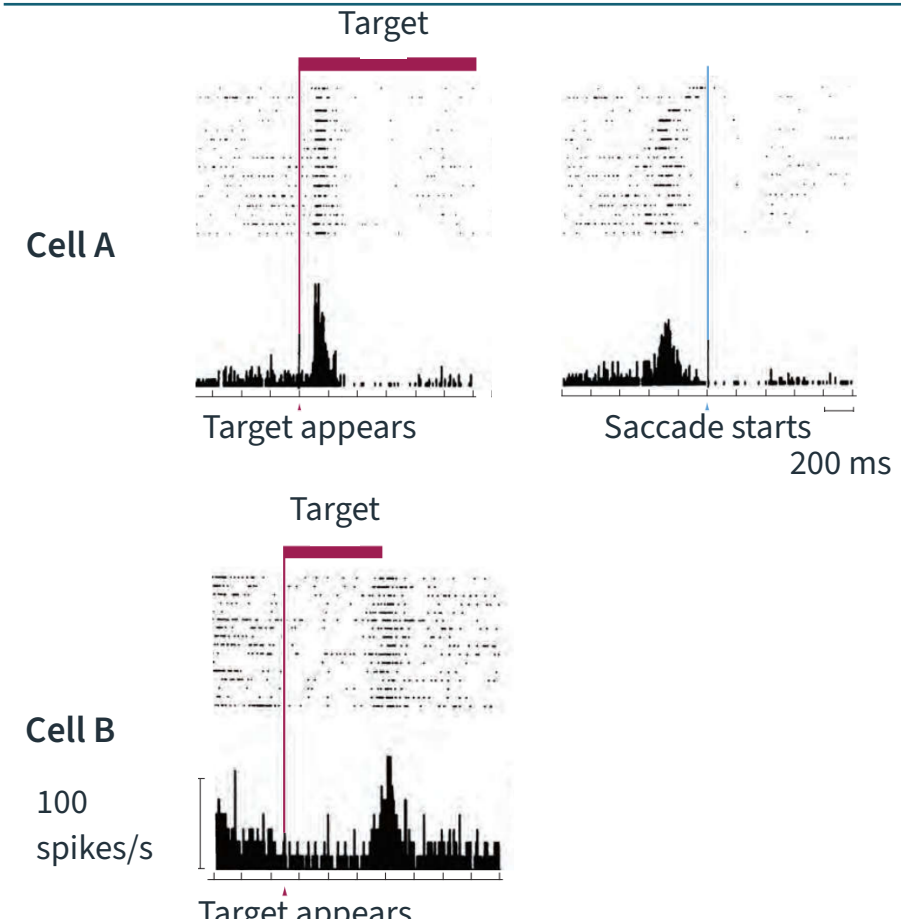


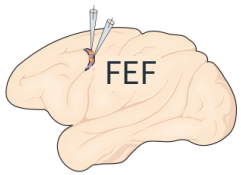


- Stimulating FEF evokes saccades

Kandel 6e, Fig. 35–13, p. 876 · two FEF neurons (Bruce & Goldberg 1985); panel headings removed, they state the answer; labels re-set from the figure

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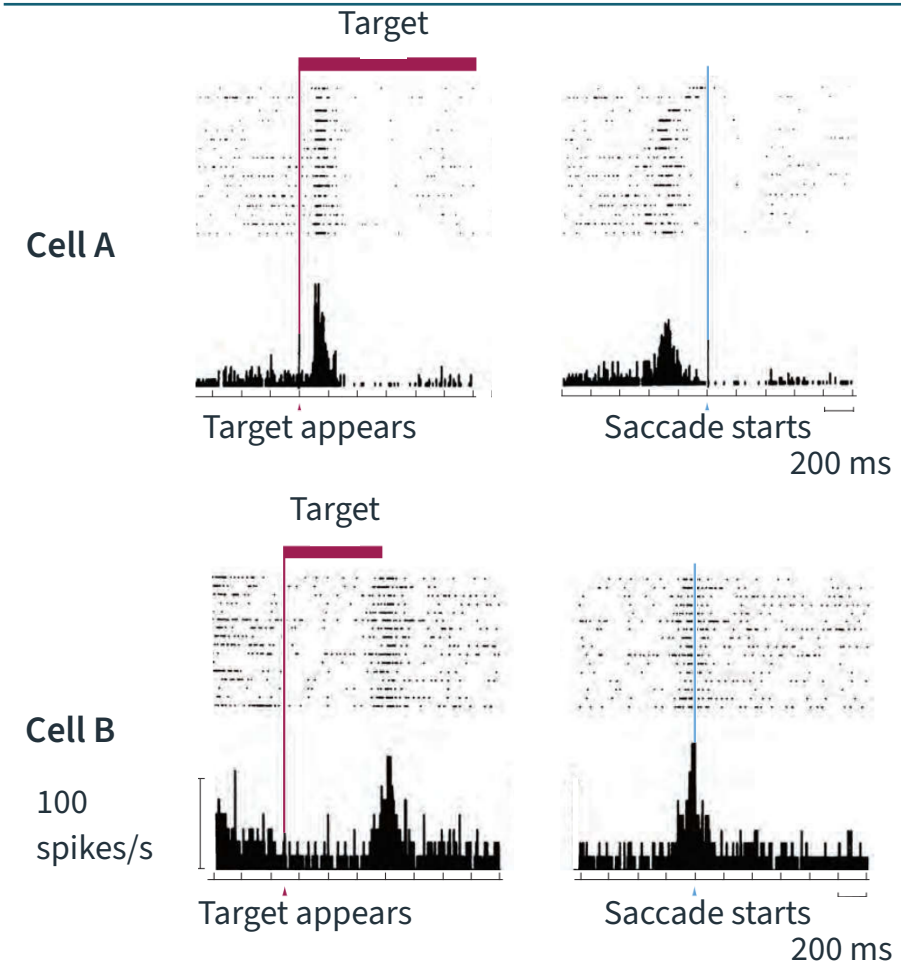


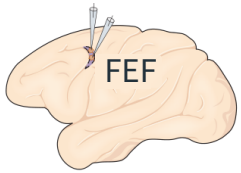


- Stimulating FEF evokes saccades

Kandel 6e, Fig. 35–13, p. 876 · two FEF neurons (Bruce & Goldberg 1985); panel headings removed, they state the answer; labels re-set from the figure

Aligning on two events shows what an FEF cell encodes





- Stimulating FEF evokes saccades

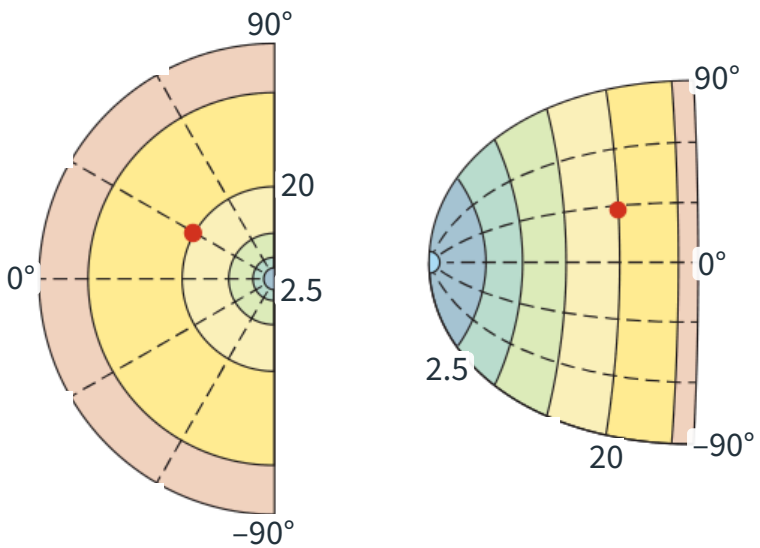
- Cell A follows the target: a visual cell.
- Cell B follows the saccade: a movement cell.

Kandel 6e, Fig. 35–13, p. 876 · two FEF neurons (Bruce & Goldberg 1985); panel headings removed, they state the answer; labels re-set from the figure

The colliculus maps the visual field

Left visual field

Right colliculus map

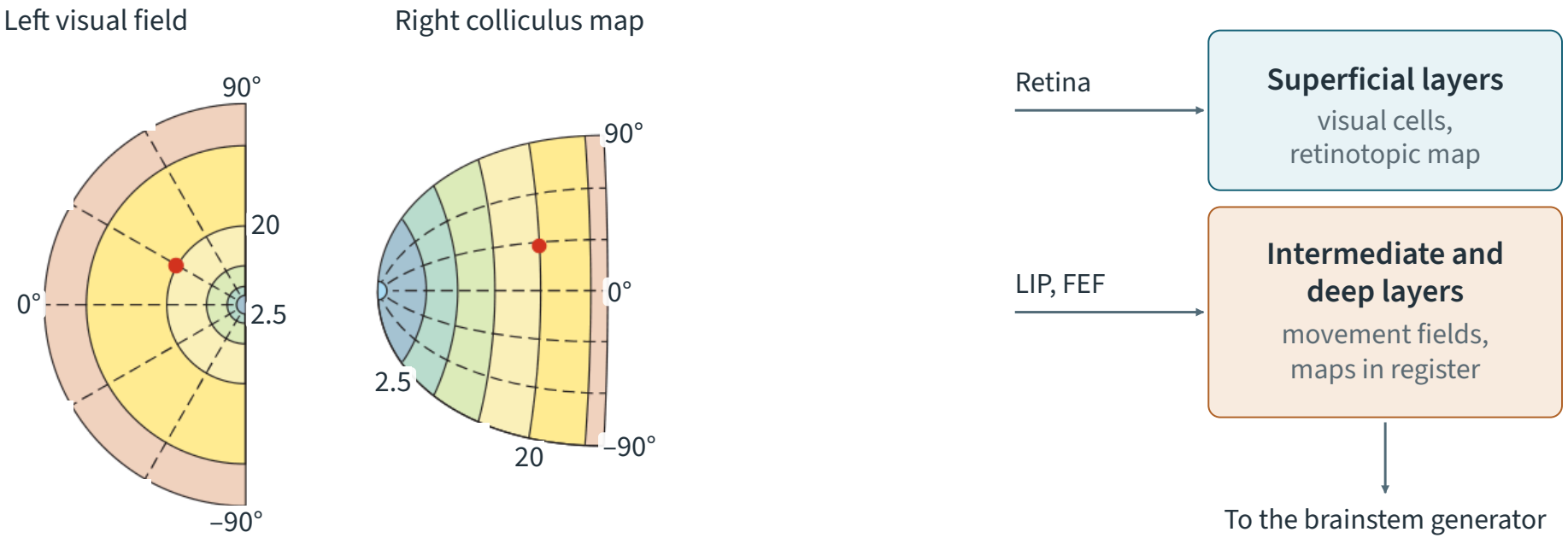


Retina → **Superficial layers**
visual cells,
retinotopic map

Kandel 6e, Fig. 35–10, p. 872 · visual field and collicular map in polar coordinates (Quaia et al. 1998), a subset of labels re-set; layers: course schematic after pp. 871–873

- Each place in the field has its own place on the map.

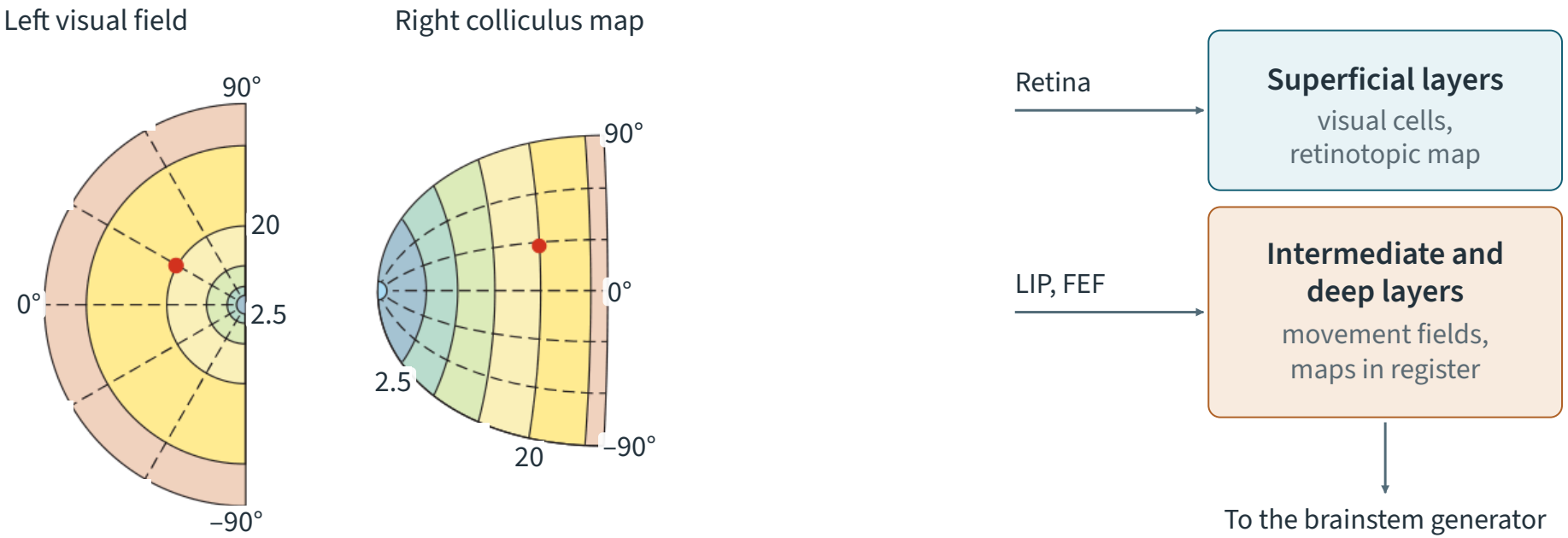
The colliculus maps the visual field



Kandel 6e, Fig. 35–10, p. 872 · visual field and collicular map in polar coordinates (Quaia et al. 1998), a subset of labels re-set; layers: course schematic after pp. 871–873

- Each place in the field has its own place on the map.
- Visual and movement maps lie in register: a place on the map **is** a saccade vector.

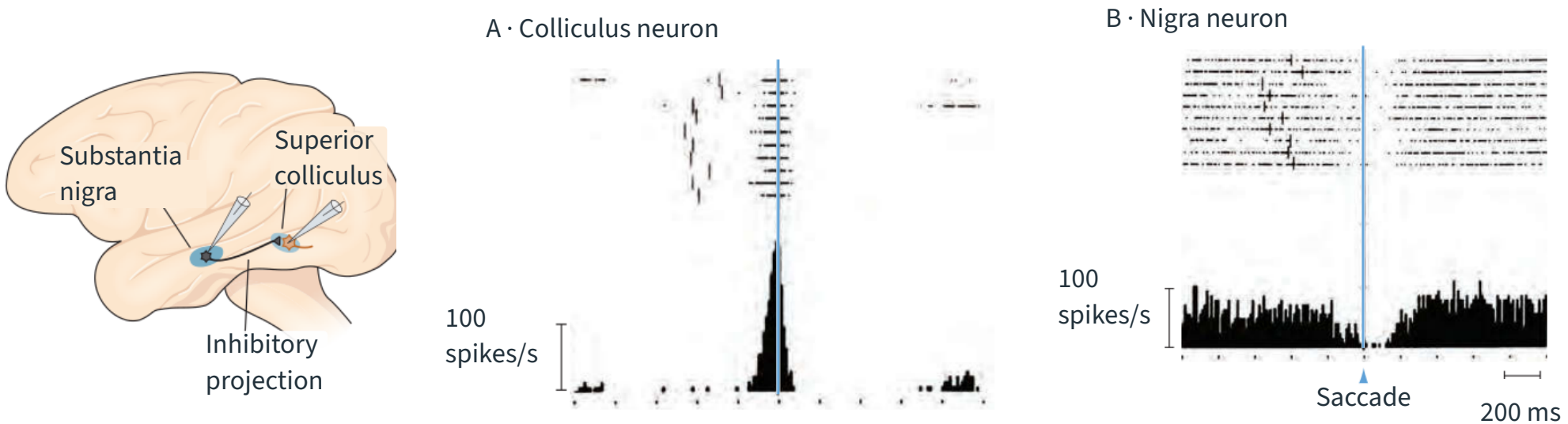
The colliculus maps the visual field



Kandel 6e, Fig. 35–10, p. 872 · visual field and collicular map in polar coordinates (Quaia et al. 1998), a subset of labels re-set; layers: course schematic after pp. 871–873

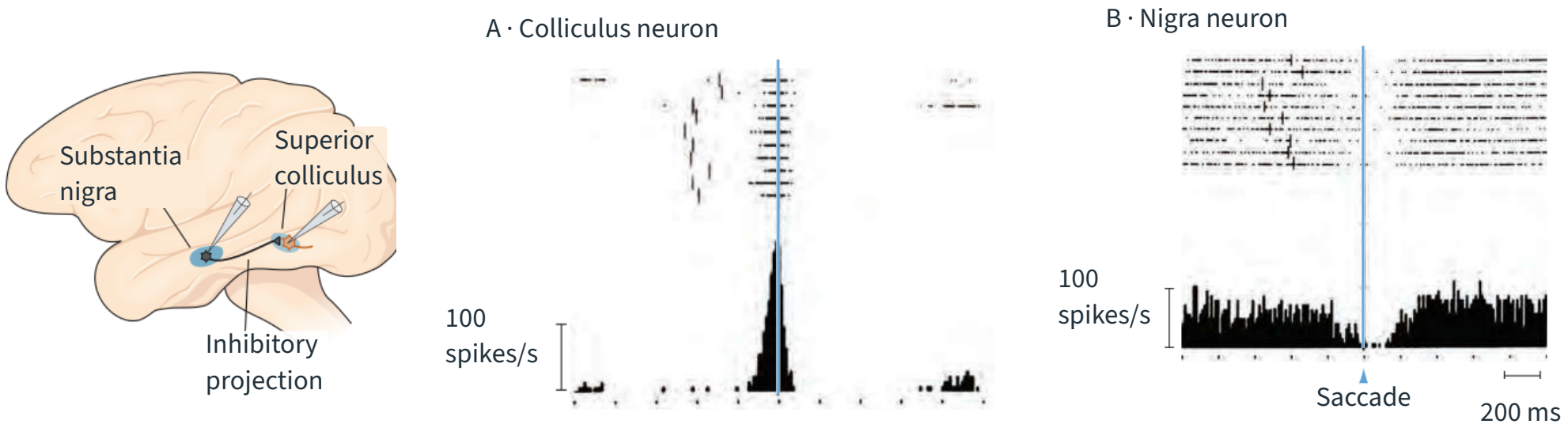
- Each place in the field has its own place on the map.
- Visual and movement maps lie in register: a place on the map **is** a saccade vector.
- A reach must relate target and hand; a saccade needs only the place.

The basal ganglia gate the colliculus



Kandel 6e, Fig. 35–11, p. 873 · superior colliculus and substantia nigra (pars reticulata) neurons around a saccade (Hikosaka & Wurtz); panels set side by side, labels re-set

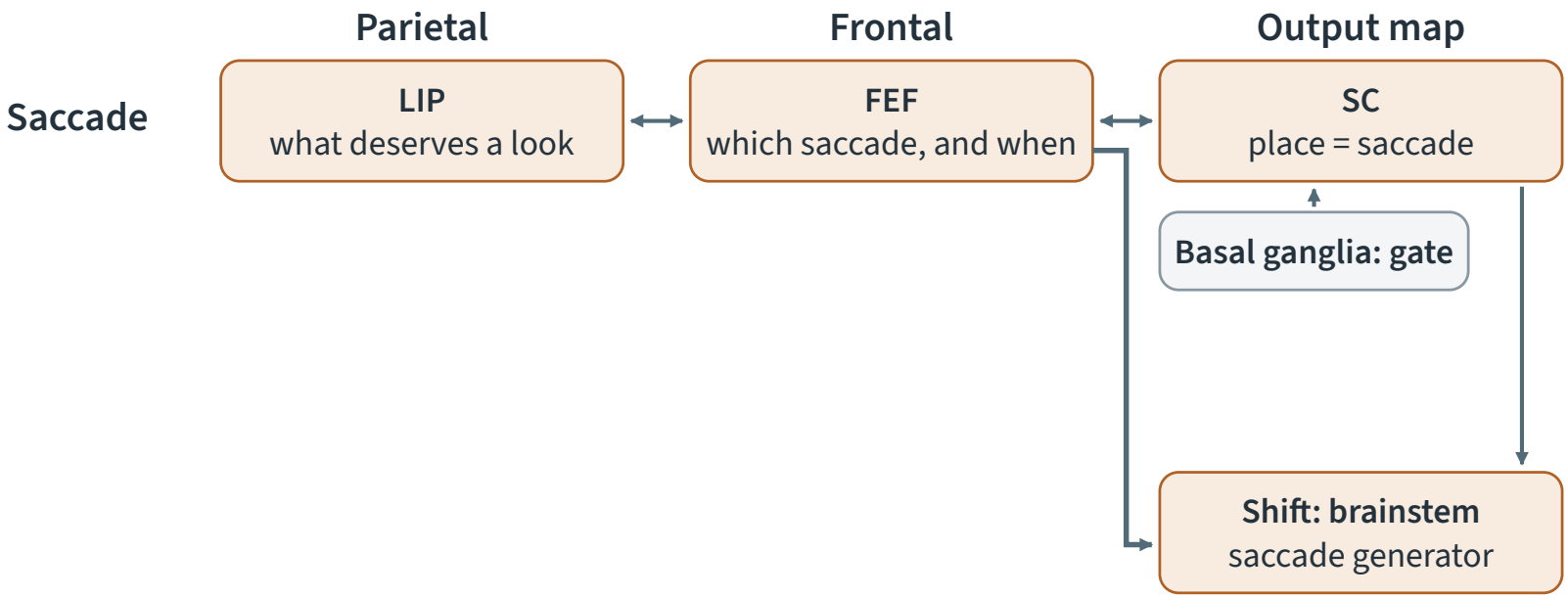
The basal ganglia gate the colliculus



Kandel 6e, Fig. 35–11, p. 873 · superior colliculus and substantia nigra (pars reticulata) neurons around a saccade (Hikosaka & Wurtz); panels set side by side, labels re-set

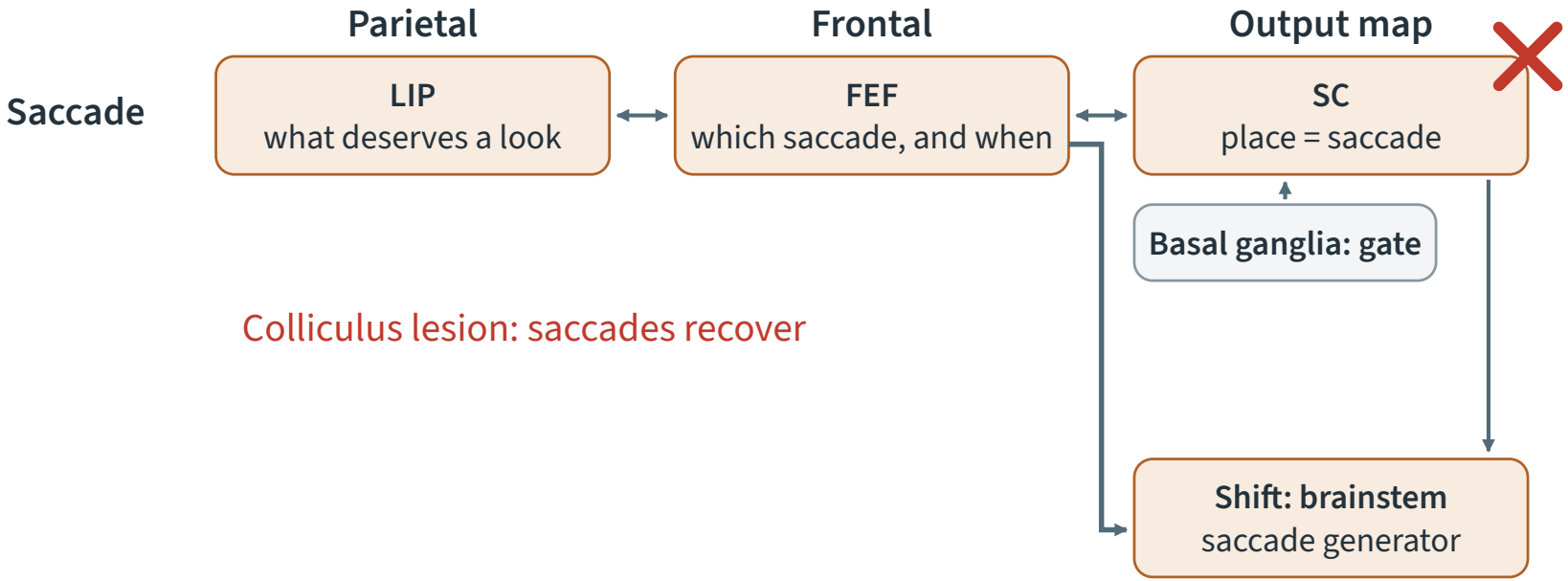
- The nigra pauses as the colliculus bursts: the basal ganglia gate it by disinhibition.

FEF reaches the brainstem without the colliculus



Kandel 6e, Ch. 35, pp. 873, 875 · lecture organizer with FEF's own projection to the brainstem reticular formation; lesion outcomes in monkeys

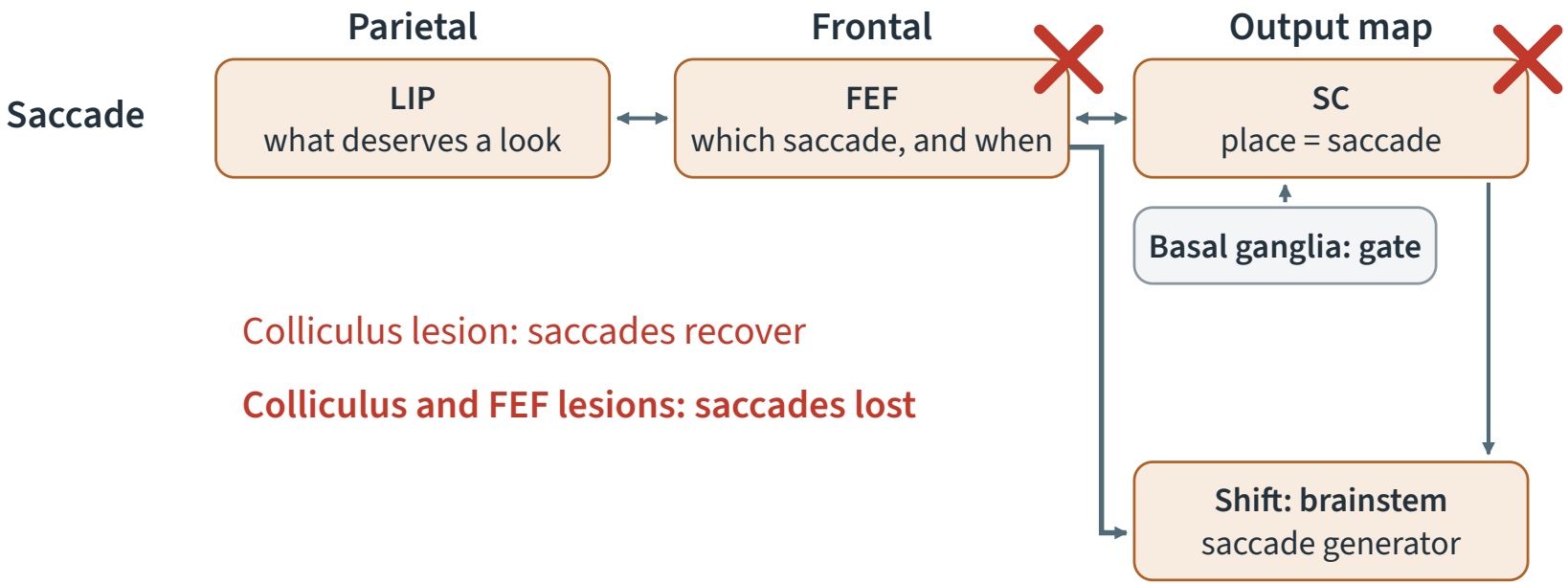
FEF reaches the brainstem without the colliculus



Kandel 6e, Ch. 35, pp. 873, 875 · lecture organizer with FEF's own projection to the brainstem reticular formation; lesion outcomes in monkeys

- FEF's route alone is enough, in time.

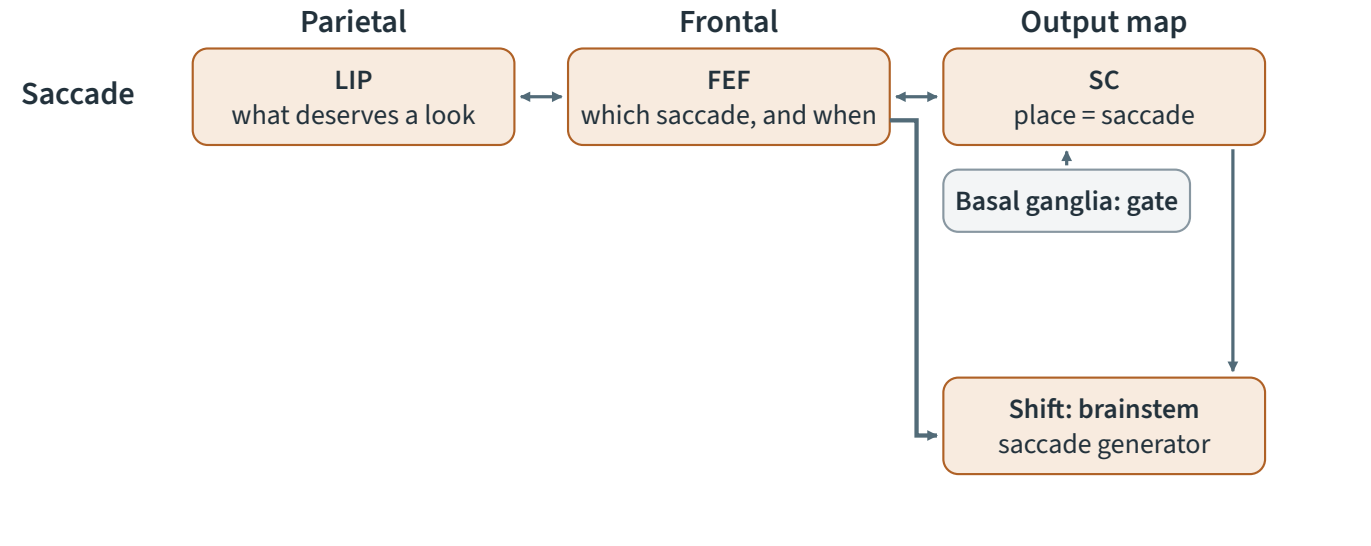
FEF reaches the brainstem without the colliculus



Kandel 6e, Ch. 35, pp. 873, 875 · lecture organizer with FEF's own projection to the brainstem reticular formation; lesion outcomes in monkeys

- FEF's route alone is enough, in time.
- Saccades need either the colliculus or FEF.

Section C summary: cortex picks the target, the colliculus moves the eye

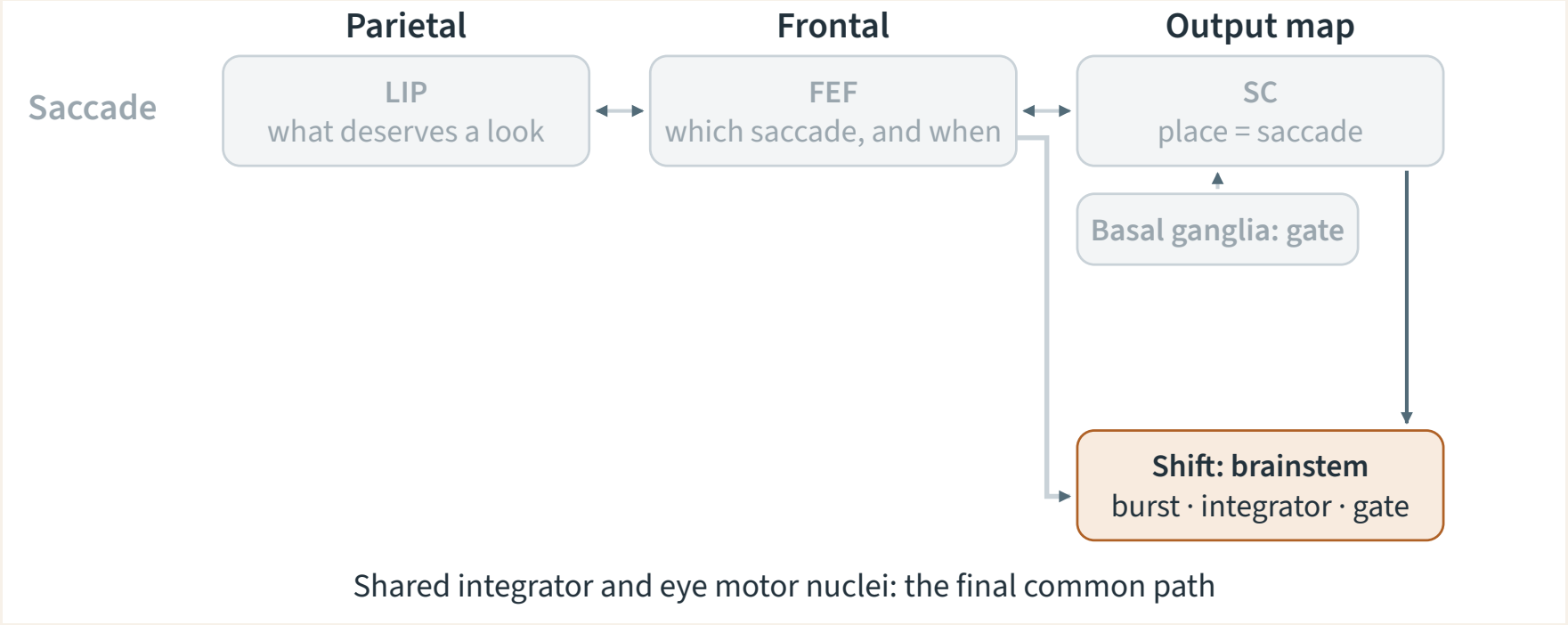


TAKE HOME

- LIP marks what deserves a look; FEF picks the saccade.
- The colliculus turns a place into a movement, gated by the basal ganglia.
- FEF also has its own route to the brainstem.

Kandel 6e, Ch. 35, pp. 871–876 · the section C map, returning, with FEF's own route

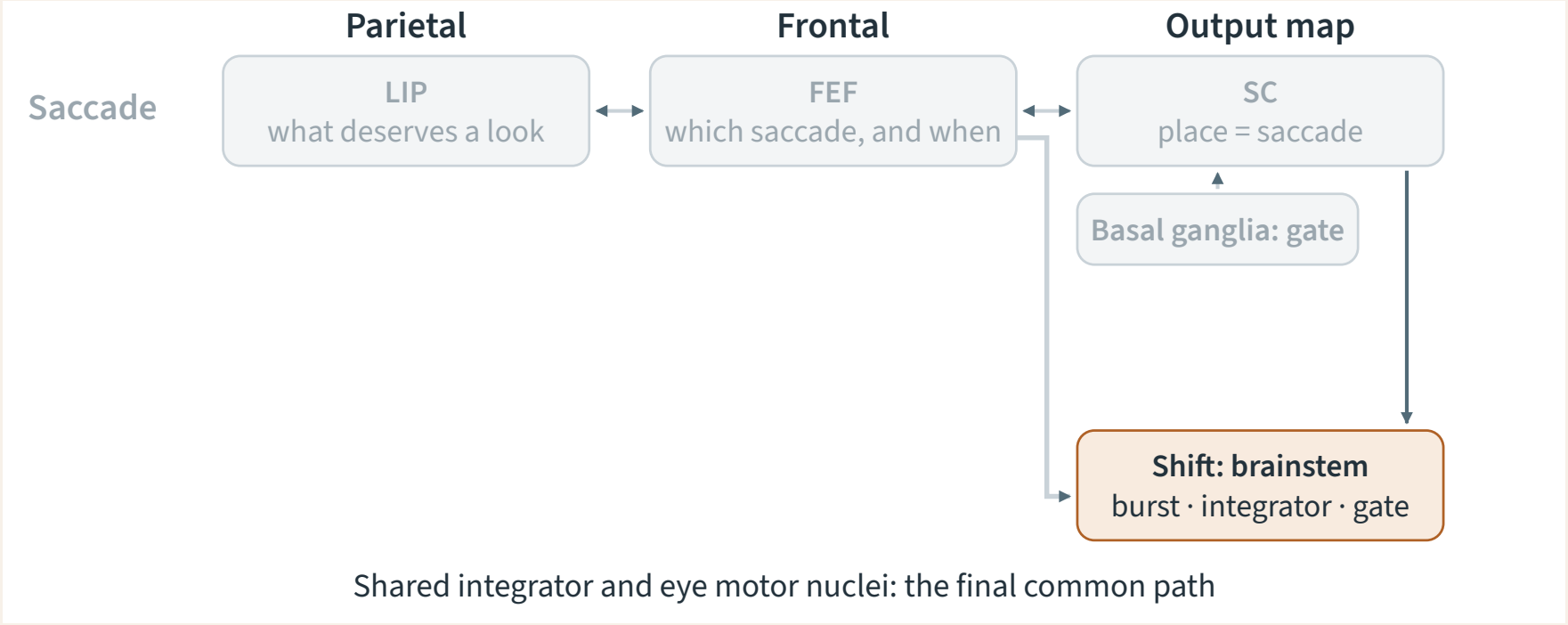
D · Saccade generation



Shared integrator and eye motor nuclei: the final common path

Kandel 6e, Ch. 35 · lecture organizer, built up section by section; a teaching schematic, not measured wiring

D · Saccade generation

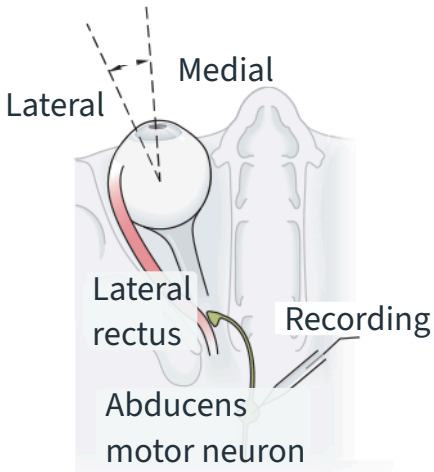


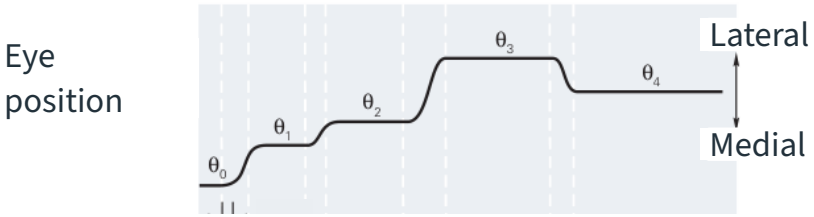
Shared integrator and eye motor nuclei: the final common path

Kandel 6e, Ch. 35 · lecture organizer, built up section by section; a teaching schematic, not measured wiring

- The brainstem turns the chosen saccade into a burst and a step, on a final common path.

Recording one abducens motor neuron

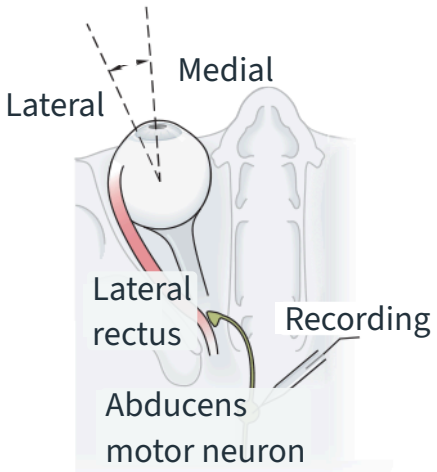


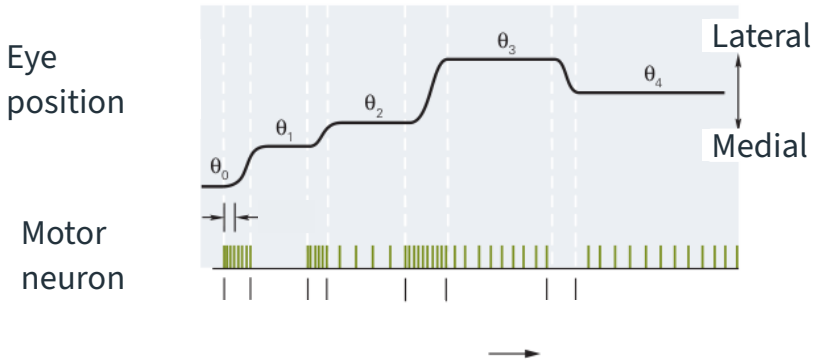


- The orbit's load needs a burst against viscosity and a step against elasticity.

Kandel 6e, Fig. 35–7A, p. 868 · abducens motor neuron in the monkey (Fuchs & Luschei 1970); labels re-set from the figure

Recording one abducens motor neuron

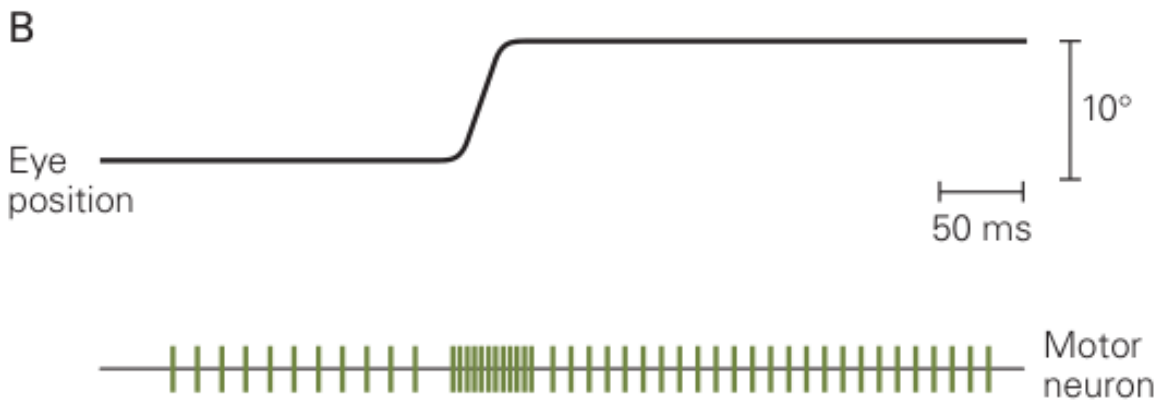




- The orbit's load needs a burst against viscosity and a step against elasticity.
- The neuron fires:
  - Burst with each saccade: the pulse
  - Higher rate at each lateral position: the step

Kandel 6e, Fig. 35–7A, p. 868 · abducens motor neuron in the monkey (Fuchs & Luschei 1970); labels re-set from the figure

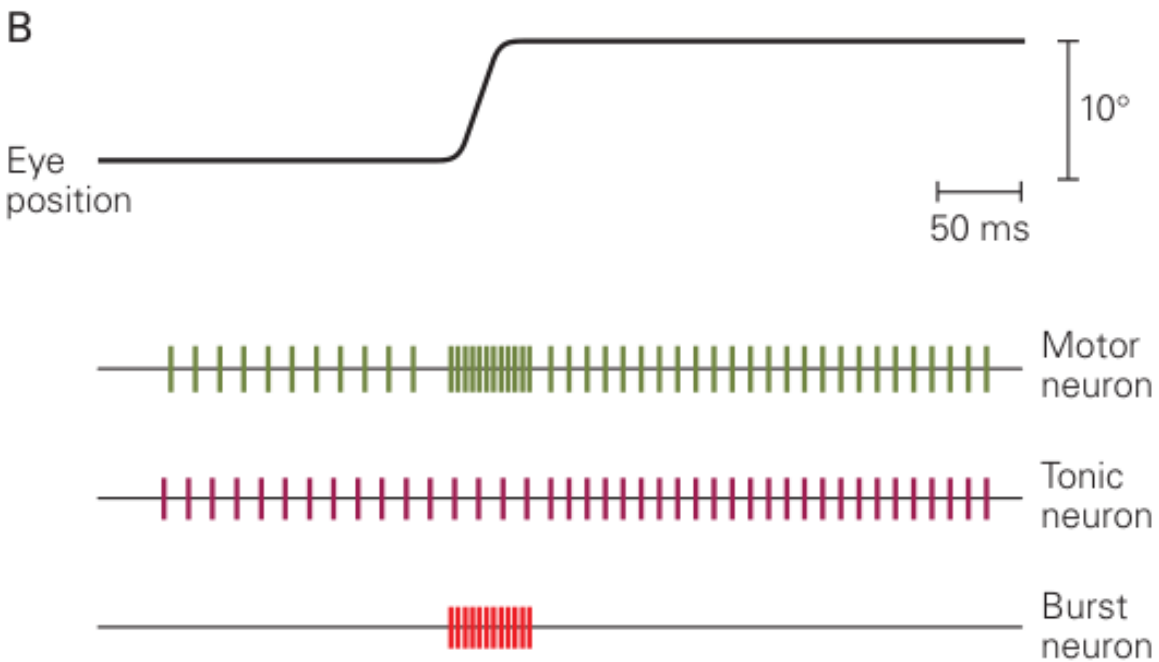
Taking the motor neuron’s signal apart, cell by cell



- **Motor neuron**: burst, then a higher steady rate

Kandel 6e, Fig. 35–8B, p. 869 · firing patterns of four brainstem cell types around a horizontal saccade

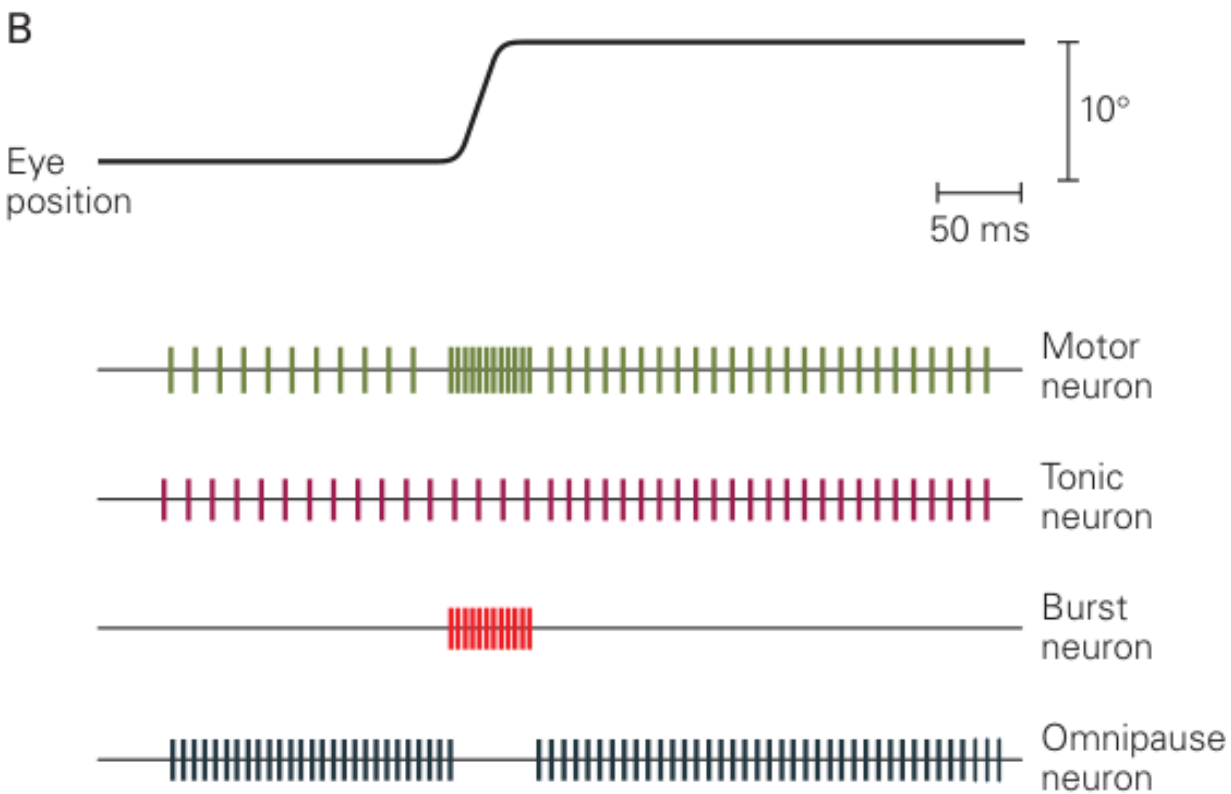
Taking the motor neuron’s signal apart, cell by cell



- **Motor neuron**: burst, then a higher steady rate
- **Tonic**: steady rate only, the position
- **Burst**: during the saccade only, the velocity

Kandel 6e, Fig. 35–8B, p. 869 · firing patterns of four brainstem cell types around a horizontal saccade

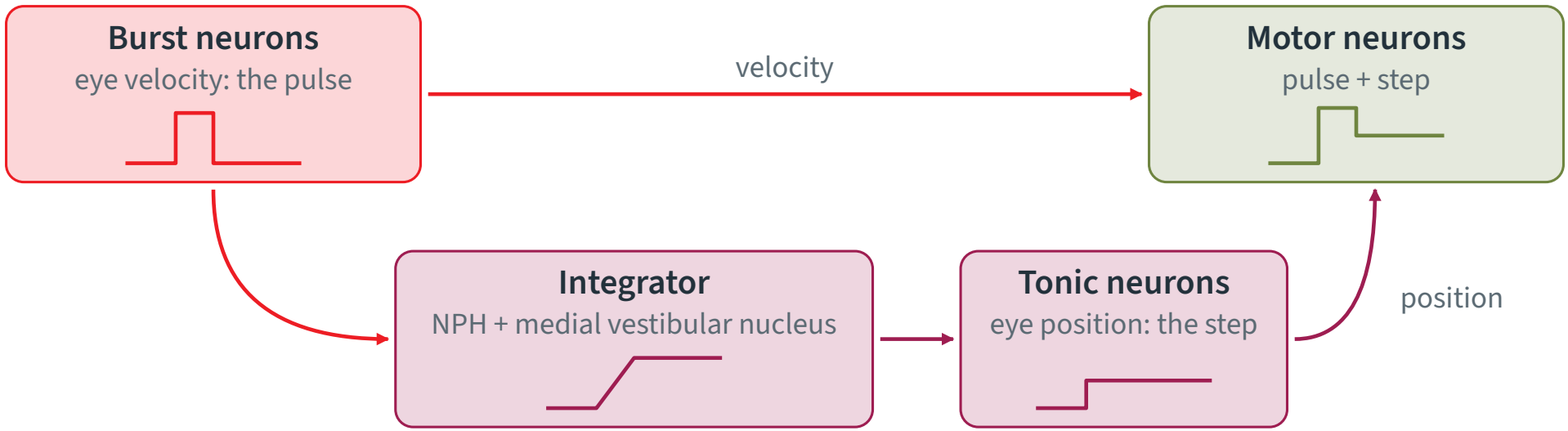
Taking the motor neuron’s signal apart, cell by cell



- **Motor neuron**: burst, then a higher steady rate
- **Tonic**: steady rate only, the position
- **Burst**: during the saccade only, the velocity
- **Omnipause**: silent only during the saccade, a gate

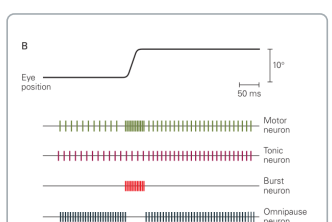
Kandel 6e, Fig. 35–8B, p. 869 · firing patterns of four brainstem cell types around a horizontal saccade

Position is the integral of velocity



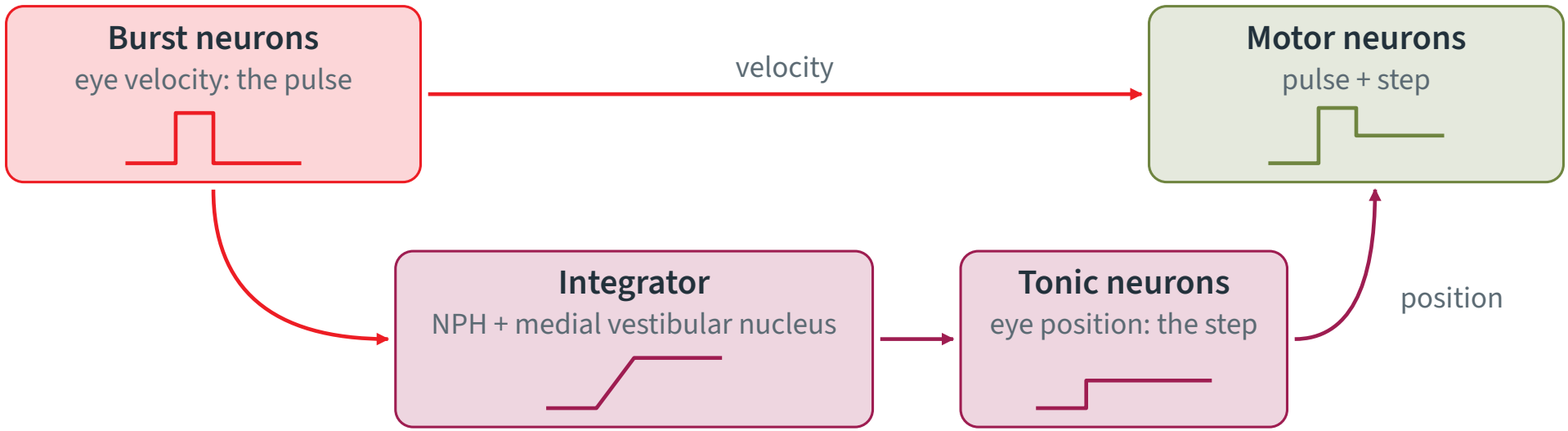
Kandel 6e, Fig. 35–8 and p. 870 · teaching simplification: the velocity signal, its integral, and their sum

- The burst drives the motor neurons and, through the integrator, sets the step.



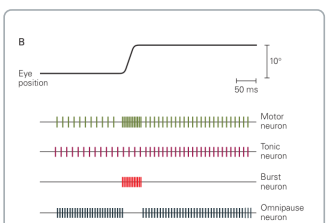
D.03 · cell types

Position is the integral of velocity



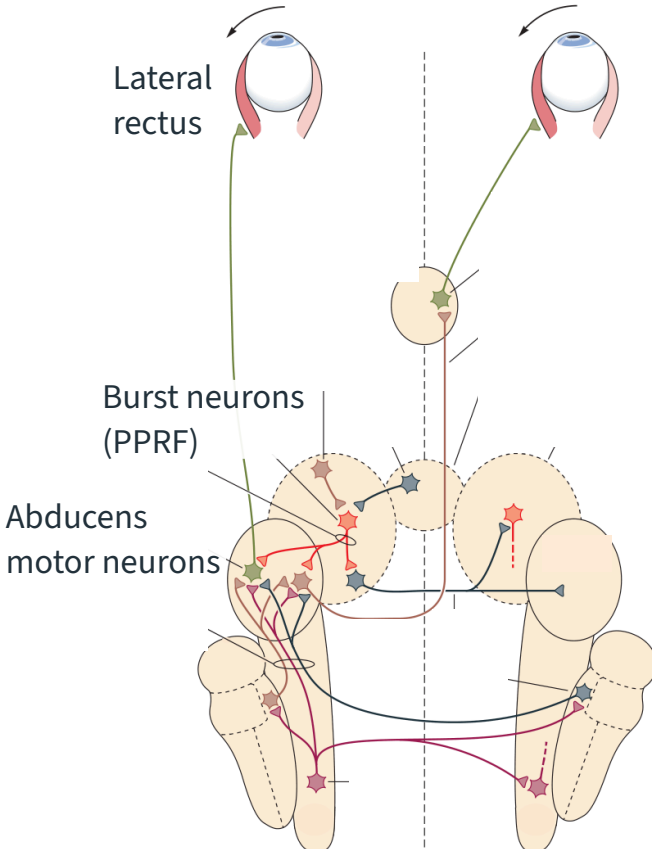
Kandel 6e, Fig. 35–8 and p. 870 · teaching simplification: the velocity signal, its integral, and their sum

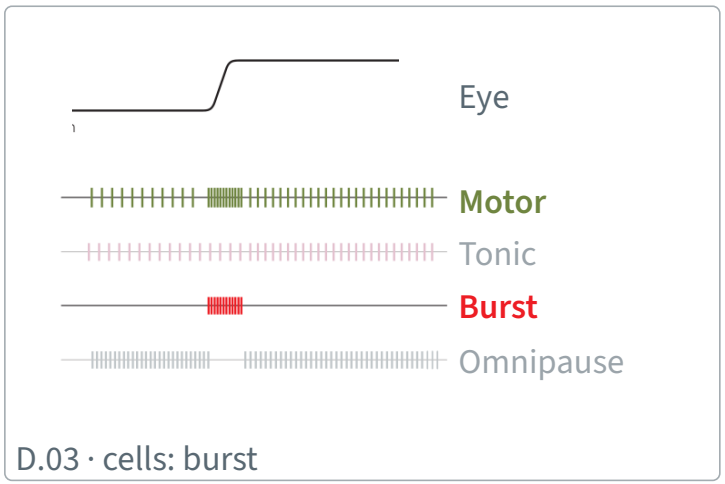
- The burst drives the motor neurons and, through the integrator, sets the step.
- Without the integrator, the eye drifts back.



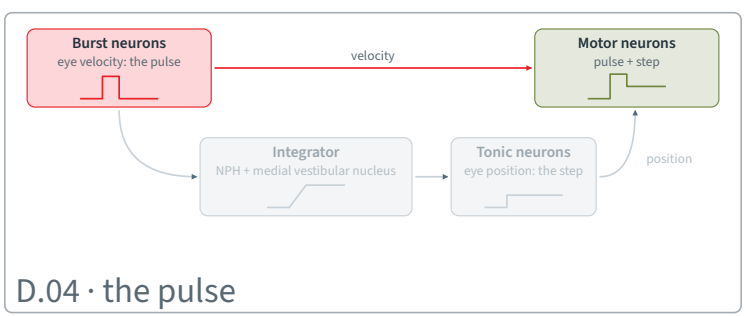
D.03 · cell types

Burst neurons make the pulse





D.03 · cells: burst

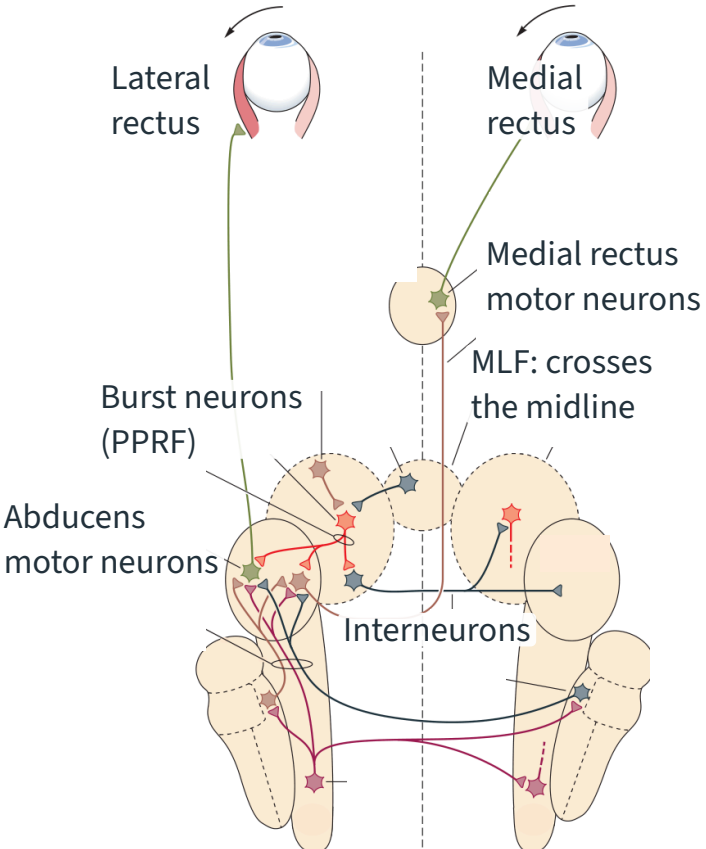


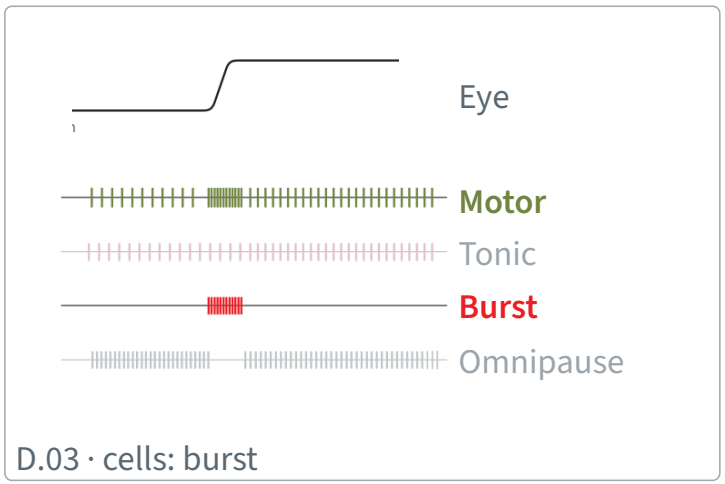
D.04 · the pulse

- Burst neurons fire the abducens motor neurons: the pulse.

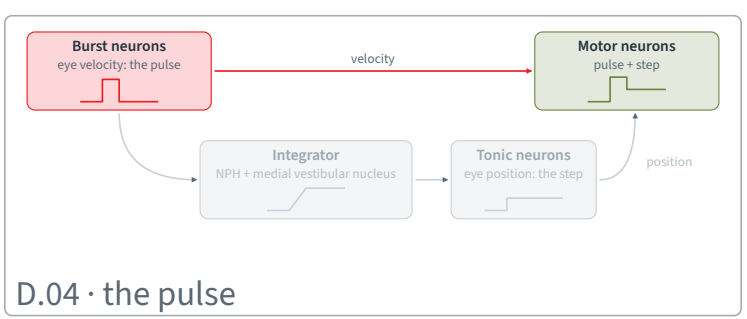
Kandel 6e, Fig. 35–8A, p. 869 and pp. 869–870 · brainstem circuit for a horizontal saccade to the left; labels re-set, others removed

Burst neurons make the pulse





D.03 · cells: burst

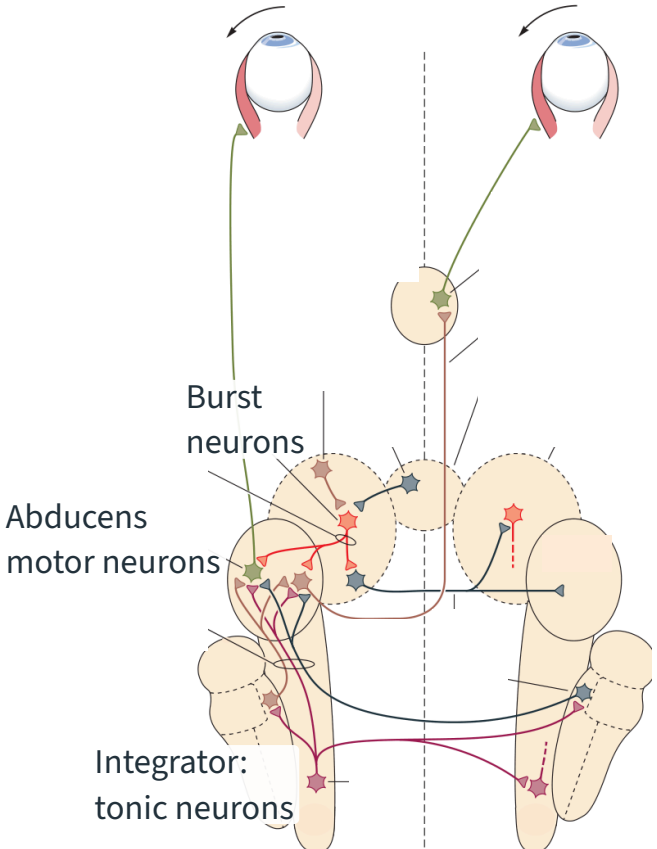


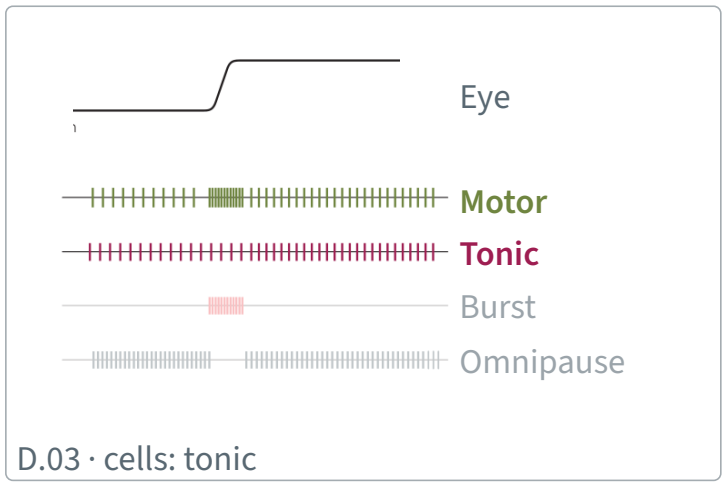
D.04 · the pulse

- Burst neurons fire the abducens motor neurons: the pulse.
- Interneurons cross to the other eye: one side moves both eyes.

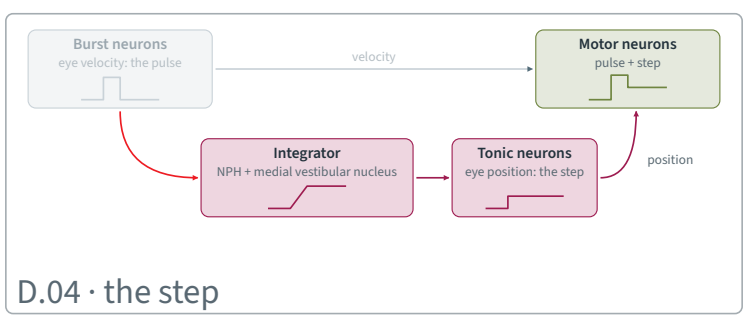
Kandel 6e, Fig. 35–8A, p. 869 and pp. 869–870 · brainstem circuit for a horizontal saccade to the left; labels re-set, others removed

An integrator makes the step





D.03 · cells: tonic

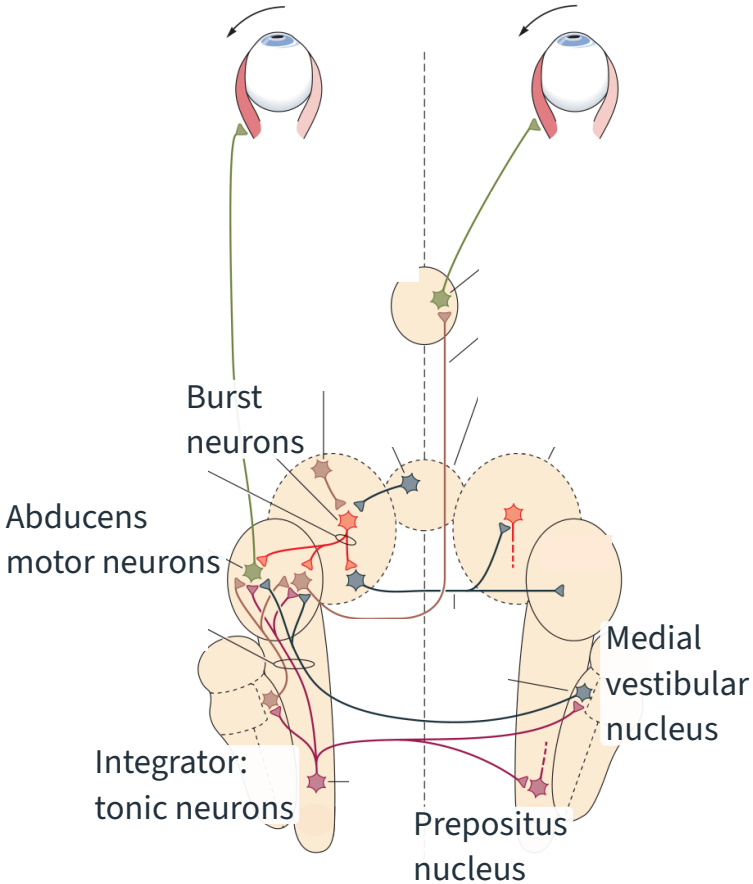


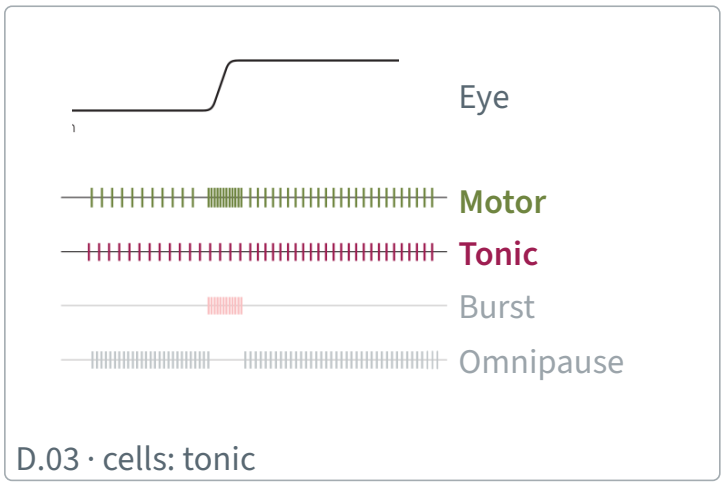
D.04 · the step

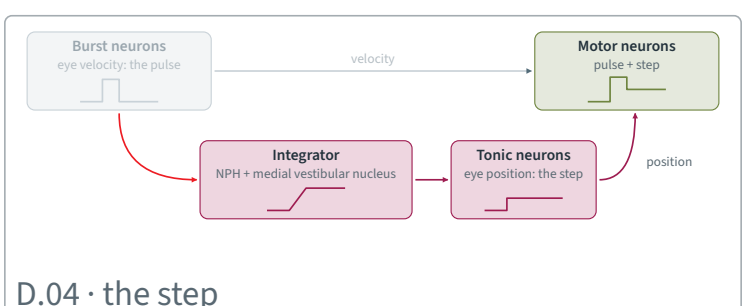
- Tonic neurons integrate the burst into a steady rate: the step.

Kandel 6e, Fig. 35–8A, p. 869 and pp. 869–870 · brainstem circuit for a horizontal saccade to the left; labels re-set, others removed

An integrator makes the step



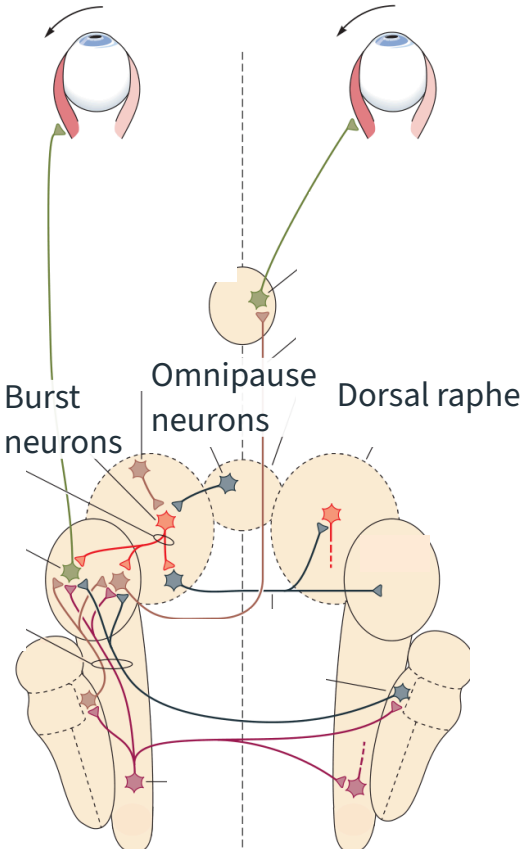


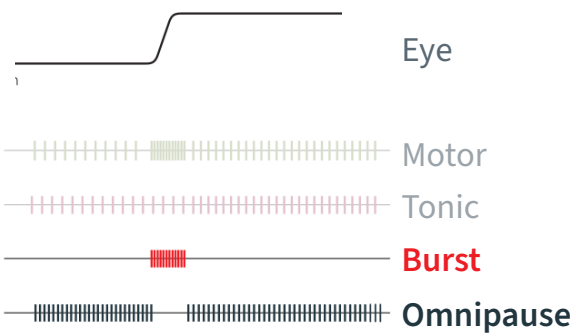


- Tonic neurons integrate the burst into a steady rate: the step.
- They lie in the medial vestibular and prepositus nuclei.

Kandel 6e, Fig. 35–8A, p. 869 and pp. 869–870 · brainstem circuit for a horizontal saccade to the left; labels re-set, others removed

Omnipause neurons gate the burst





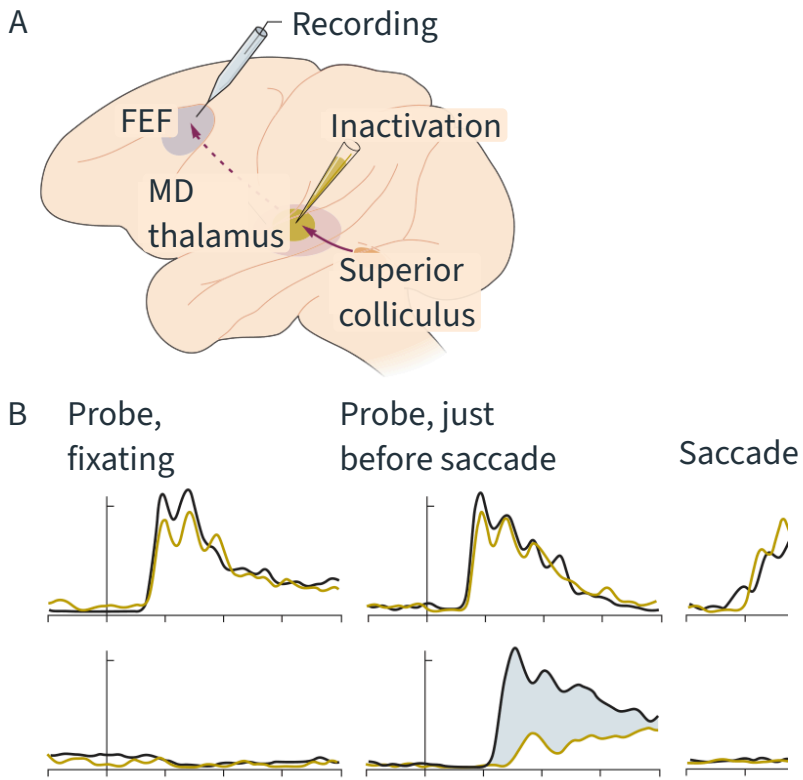
D.03 · cells: omnipause

- Omnipause neurons inhibit the burst neurons between saccades.
- They fall silent before every saccade: the gate.

Kandel 6e, Fig. 35–8A, p. 869 and pp. 869–870 · brainstem circuit for a horizontal saccade to the left; labels re-set, others removed

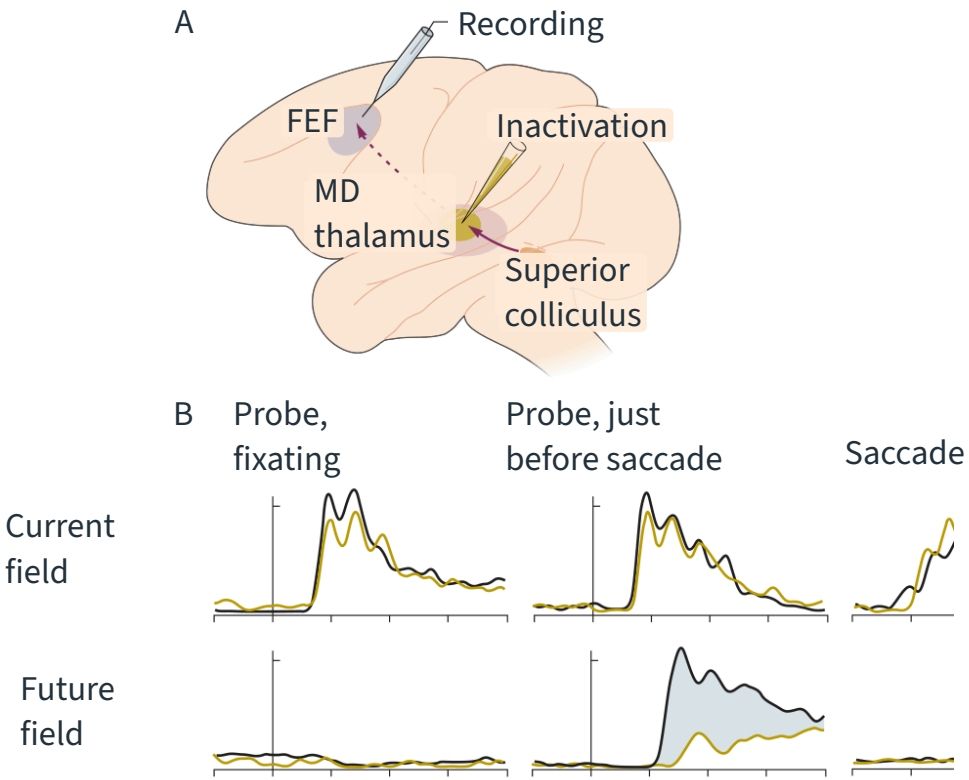
FEF receives a copy of the saccade command

- A copy of the collicular command reaches FEF through the thalamus.



Kandel 5th ed., Fig. 29–11, p. 647 · corollary discharge from colliculus through thalamus to FEF (Sommer & Wurtz); Kandel 6e, Ch. 35, p. 877; labels re-set from the figure

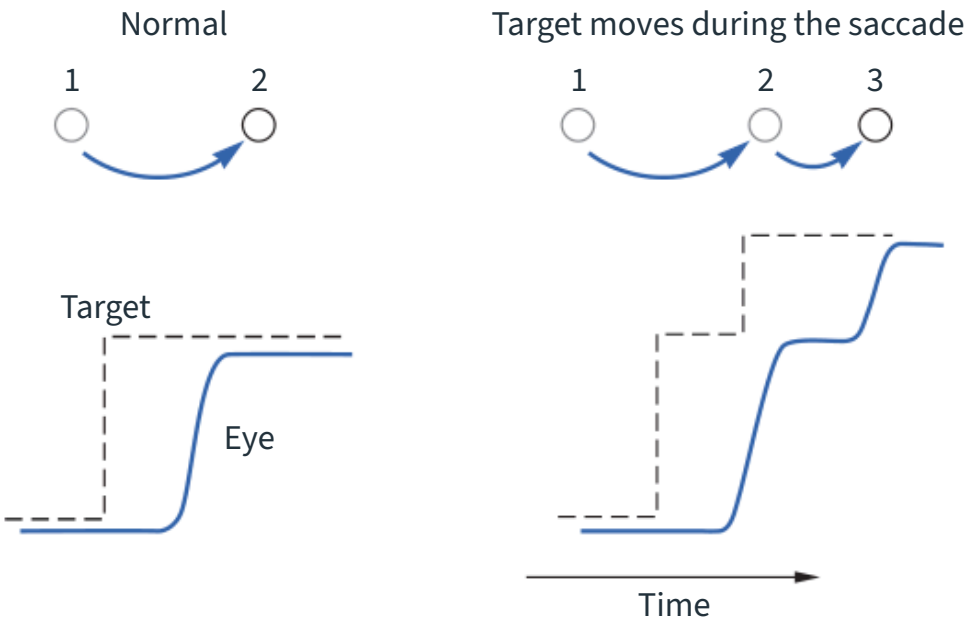
FEF receives a copy of the saccade command



- A copy of the collicular command reaches FEF through the thalamus.
- Blocking the relay spares the current field but cuts the shift to the future field.

Kandel 5th ed., Fig. 29–11, p. 647 · corollary discharge from colliculus through thalamus to FEF (Sommer & Wurtz); Kandel 6e, Ch. 35, p. 877; labels re-set from the figure

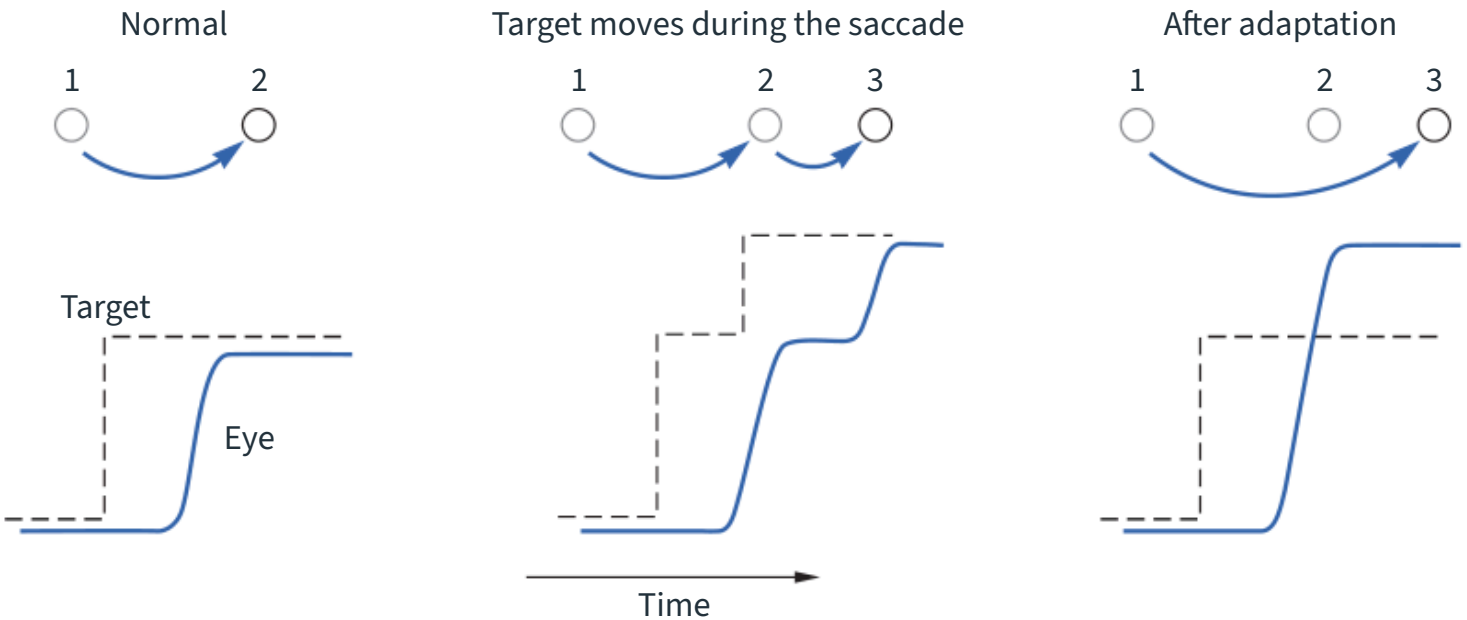
Saccades recalibrate from their own errors



Kandel 6e, Fig. 37–13B, p. 927; Ch. 37, p. 925; Ch. 35, p. 877 · saccadic adaptation to a target moved during the saccade; labels re-set

- A target moved mid-saccade leaves an error after the eye lands.

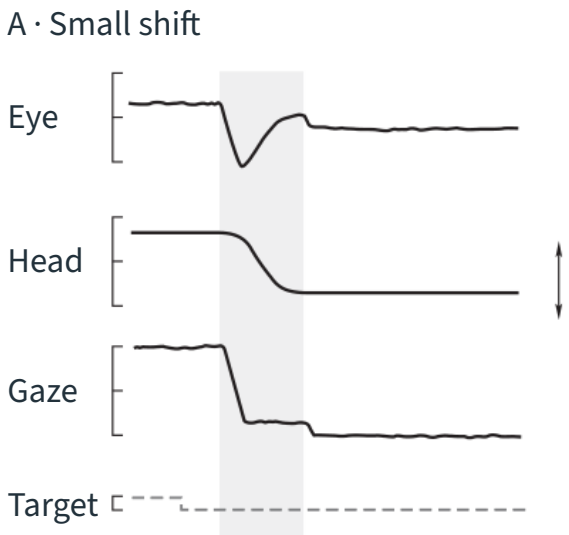
Saccades recalibrate from their own errors



Kandel 6e, Fig. 37–13B, p. 927; Ch. 37, p. 925; Ch. 35, p. 877 · saccadic adaptation to a target moved during the saccade; labels re-set

- A target moved mid-saccade leaves an error after the eye lands.
- Over hundreds of trials the first saccade grows: the cerebellum uses the error to recalibrate the next one.

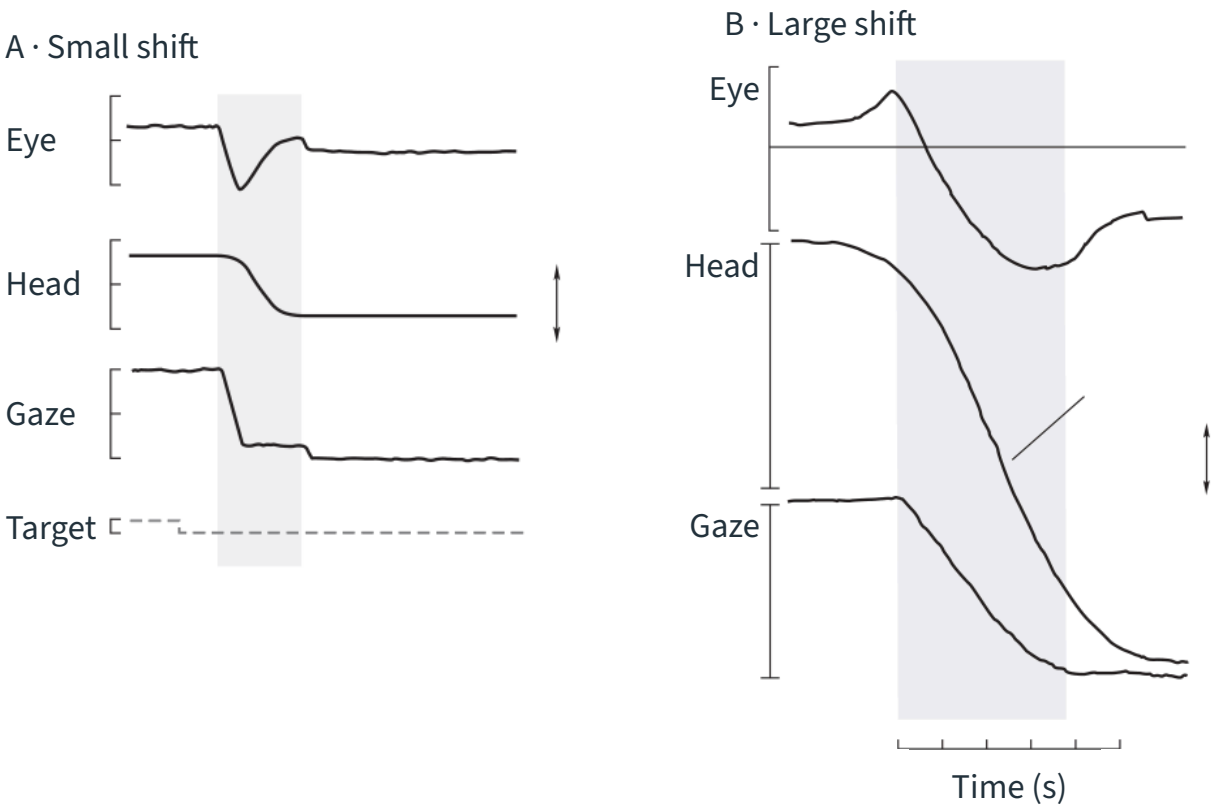
A gaze shift is a saccade plus a head turn



- Small shift: the saccade first, then the head; the reflex re-centres the eye.

Kandel 6e, Fig. 35–14 and p. 877 · eye, head and gaze during a small and a large gaze shift; panels set side by side, angle and speed scales omitted, labels re-set

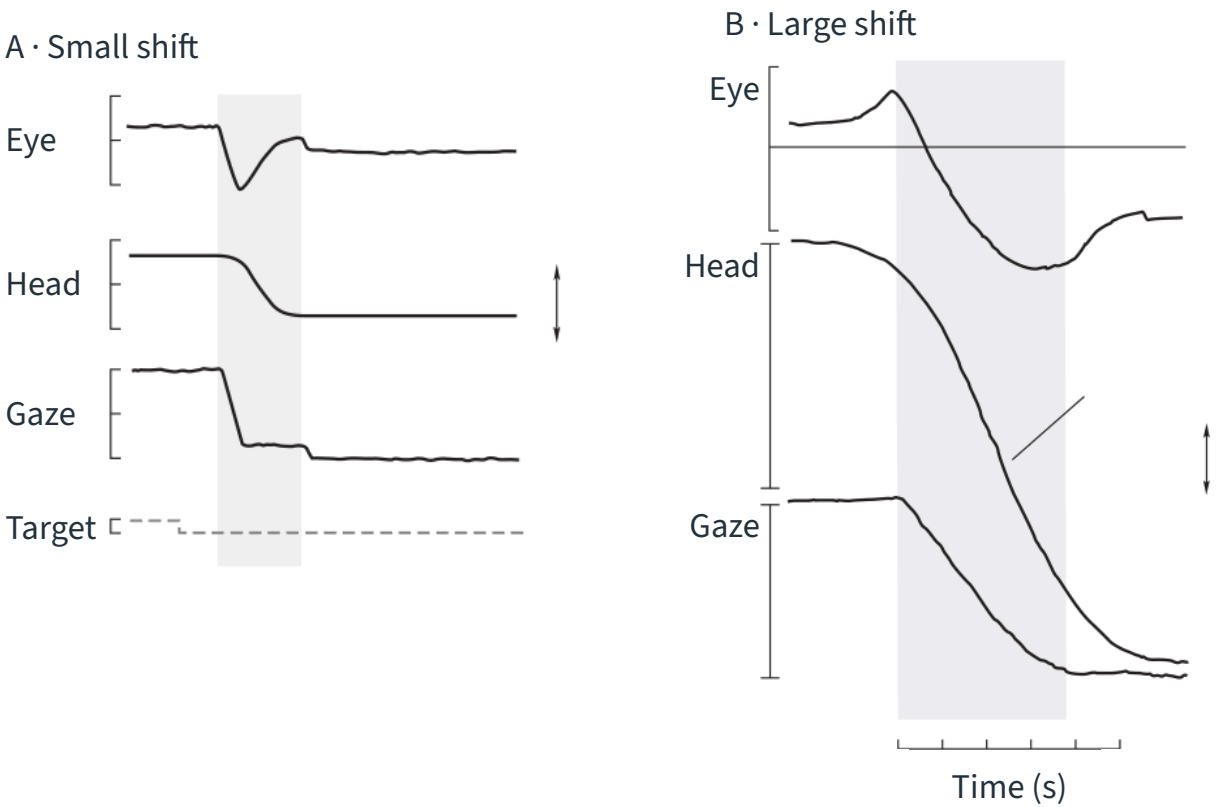
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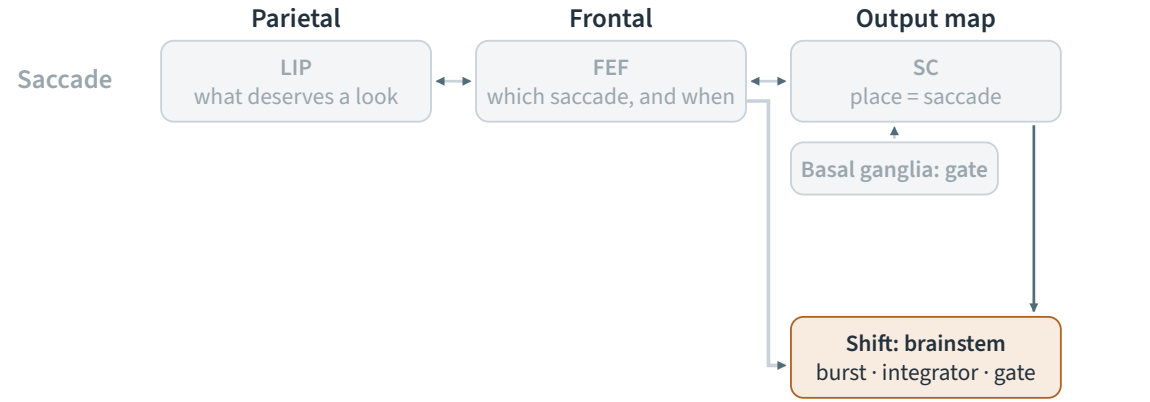
A gaze shift is a saccade plus a head turn



- Small shift: the saccade first, then the head; the reflex re-centres the eye.
- Large shift: eye and head together; the reflex is held off until the end.
- Eye and head are coordinated to move gaze.

Kandel 6e, Fig. 35–14 and p. 877 · eye, head and gaze during a small and a large gaze shift; panels set side by side, angle and speed scales omitted, labels re-set

Section D summary: the saccade command is built for the load



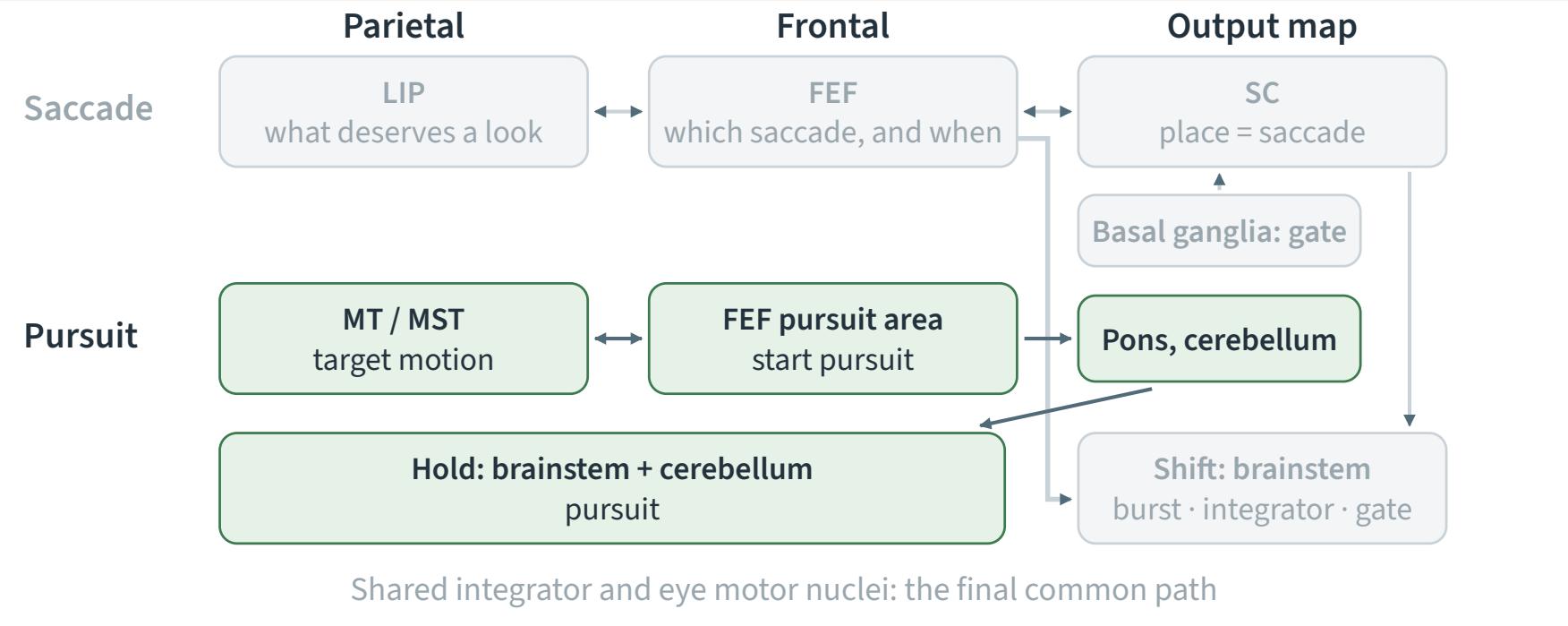
Shared integrator and eye motor nuclei: the final common path

TAKE HOME

- Pulse: burst neurons · Step: integrator · Gate: omnipause neurons
- One side of the brainstem moves both eyes.
- Errors after the saccade recalibrate the next one.
- Large gaze shifts add the head, with the reflex held off.

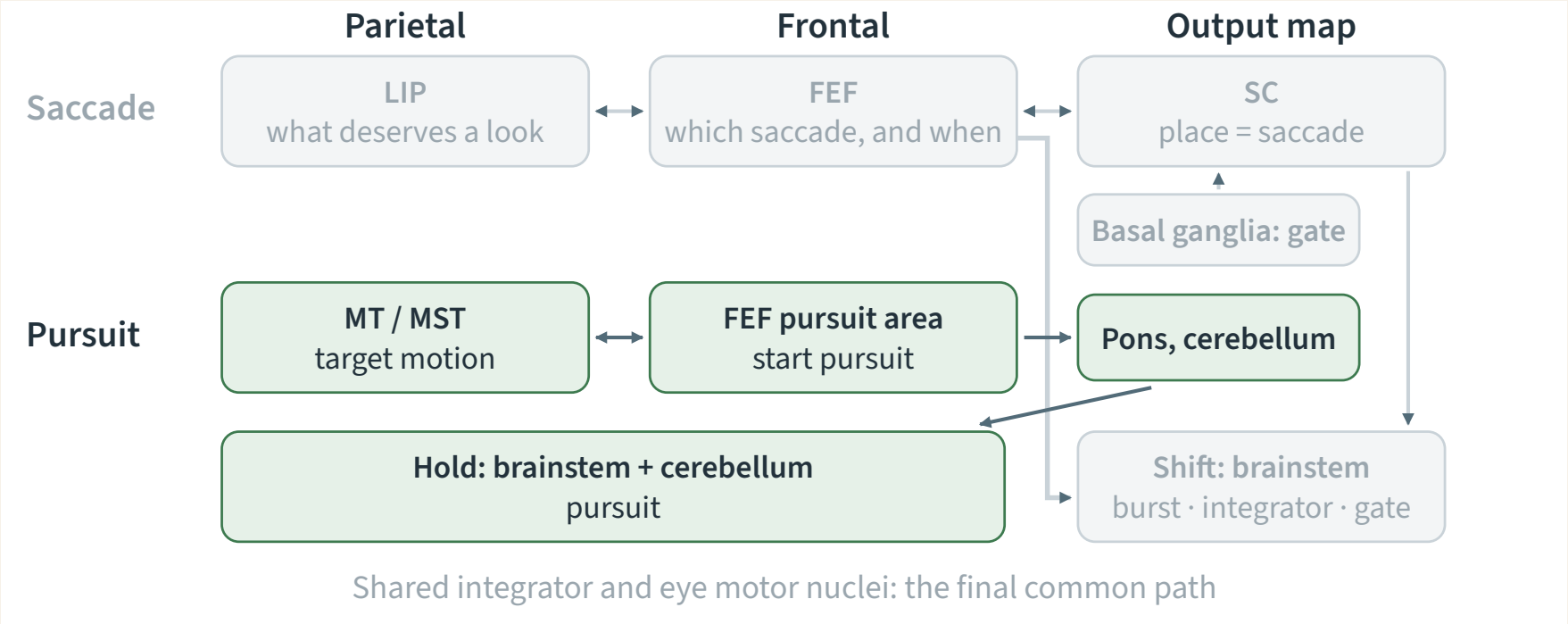
Kandel 6e, Ch. 35, pp. 868–870, 877; Ch. 37, p. 925 · the section D map, returning

E · Smooth pursuit



Kandel 6e, Ch. 35 · lecture organizer, built up section by section; a teaching schematic, not measured wiring

E · Smooth pursuit

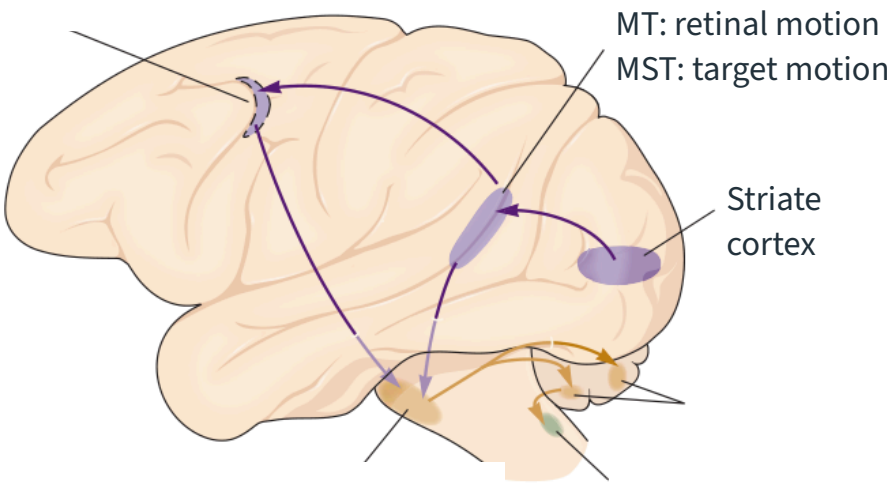


Kandel 6e, Ch. 35 · lecture organizer, built up section by section; a teaching schematic, not measured wiring

- Pursuit has its own path: motion areas and FEF, then pons and cerebellum, not the colliculus.

Pursuit bypasses the colliculus

- MT and MST sense target motion.

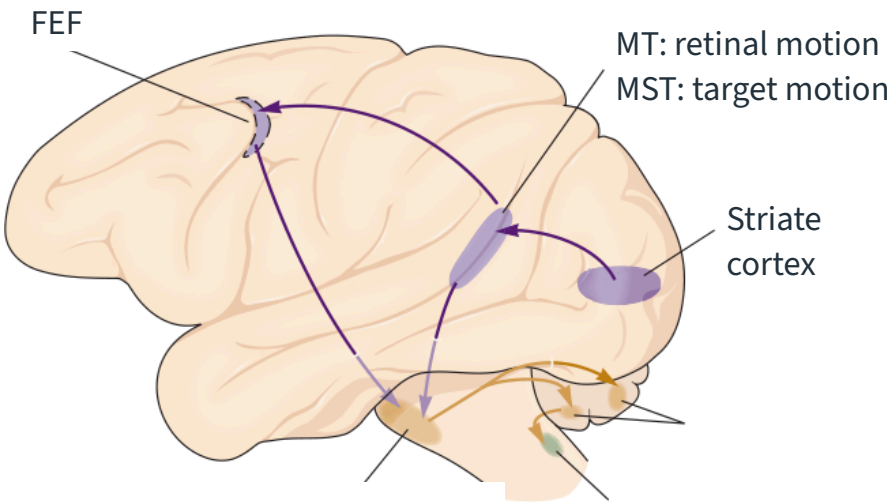


Motion processed

Kandel 6e, Fig. 35–15, p. 878 · cortical, pontine and cerebellar pathways for smooth pursuit; labels re-set and shortened from the figure

Pursuit bypasses the colliculus

- MT and MST sense target motion.
- FEF's pursuit area sets the eye velocity.

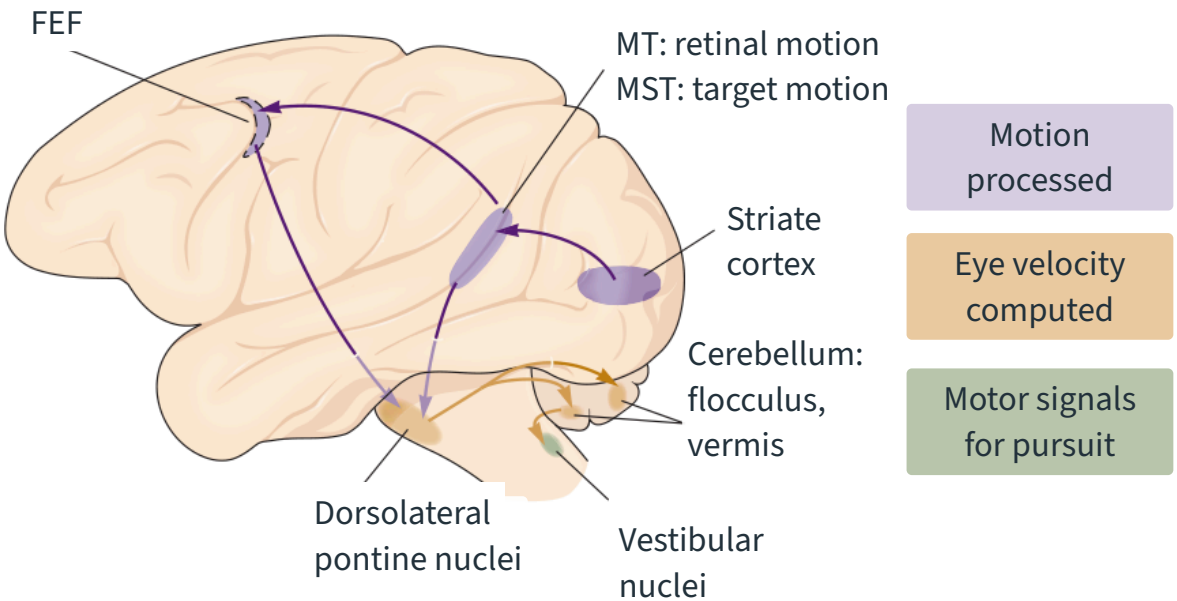


Motion processed

Eye velocity computed

Kandel 6e, Fig. 35–15, p. 878 · cortical, pontine and cerebellar pathways for smooth pursuit; labels re-set and shortened from the figure

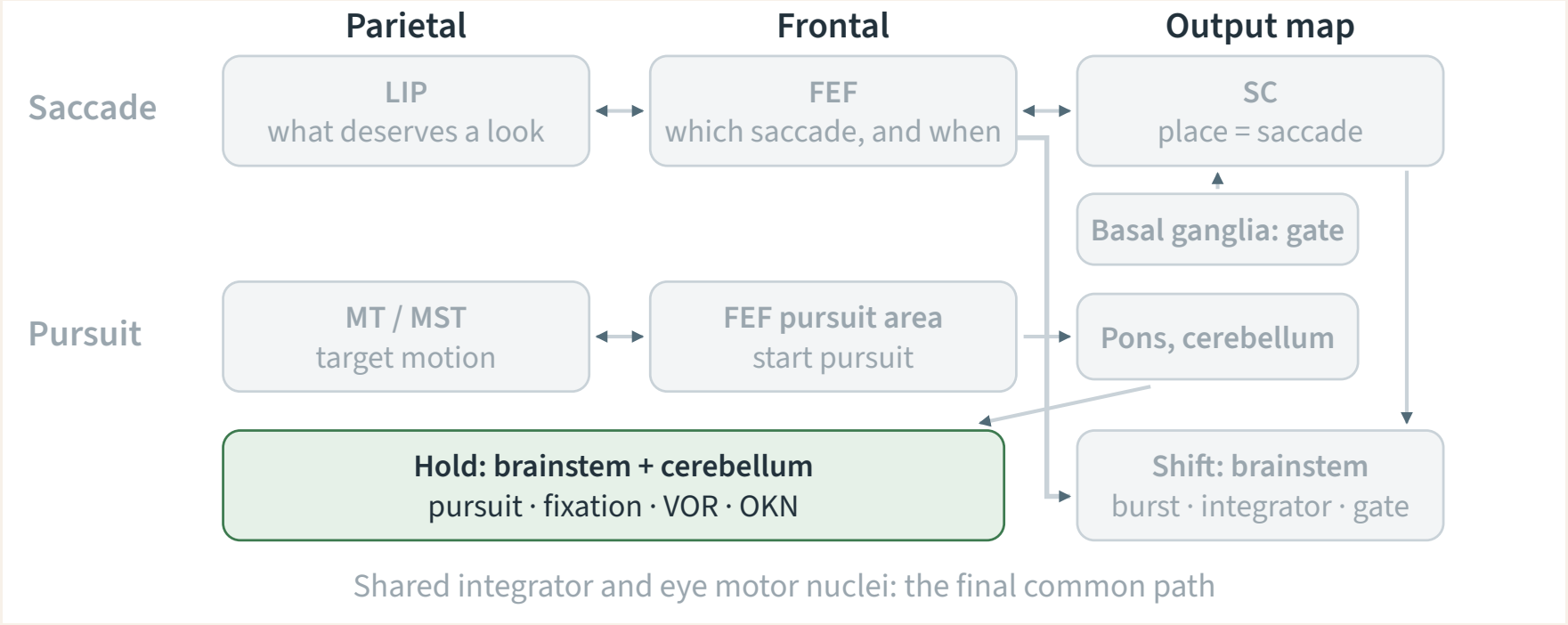
Pursuit bypasses the colliculus



- MT and MST sense target motion.
- FEF's pursuit area sets the eye velocity.
- Output: pons and cerebellum, not the colliculus.
- Like grasp beside reach, pursuit matches a **property**: velocity.

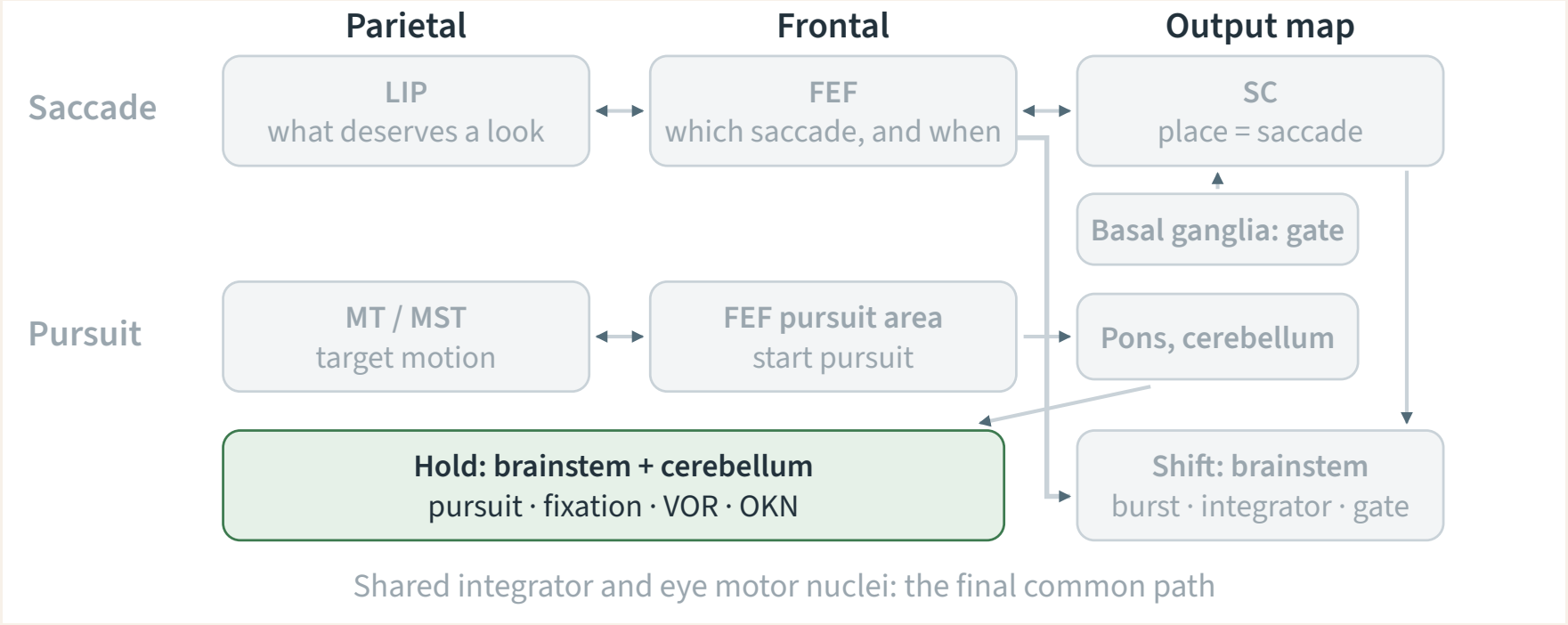
Kandel 6e, Fig. 35–15, p. 878 · cortical, pontine and cerebellar pathways for smooth pursuit; labels re-set and shortened from the figure

F · Fixation



Kandel 6e, Ch. 35 · lecture organizer, built up section by section; a teaching schematic, not measured wiring

F · Fixation



Kandel 6e, Ch. 35 · lecture organizer, built up section by section; a teaching schematic, not measured wiring

- Holding gaze: fixation and the vestibular and optokinetic reflexes, on the same brainstem and cerebellum.

Fixation is active

- Saccades are suppressed by:
  - Omnipause neurons
  - Rostral colliculus, the fixation zone
- Lesions: distractors capture the eyes

Kandel 6e, Ch. 35, pp. 869, 873; Kandel 5th ed., Ch. 29, p. 637

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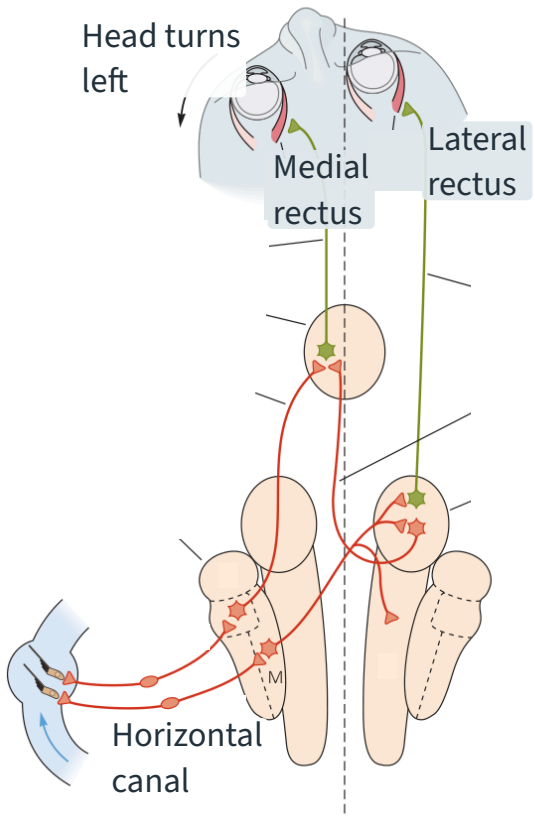
Fixation is not stillness

- Small eye movements continue
  - Without them a still image “rapidly fades to a field of gray”

> - Holding gaze on a still target is a job in its own right.
> - The other holding systems take over when the target or the head moves.

Kandel 6e, Ch. 35, pp. 869, 873; Kandel 5th ed., Ch. 29, p. 637

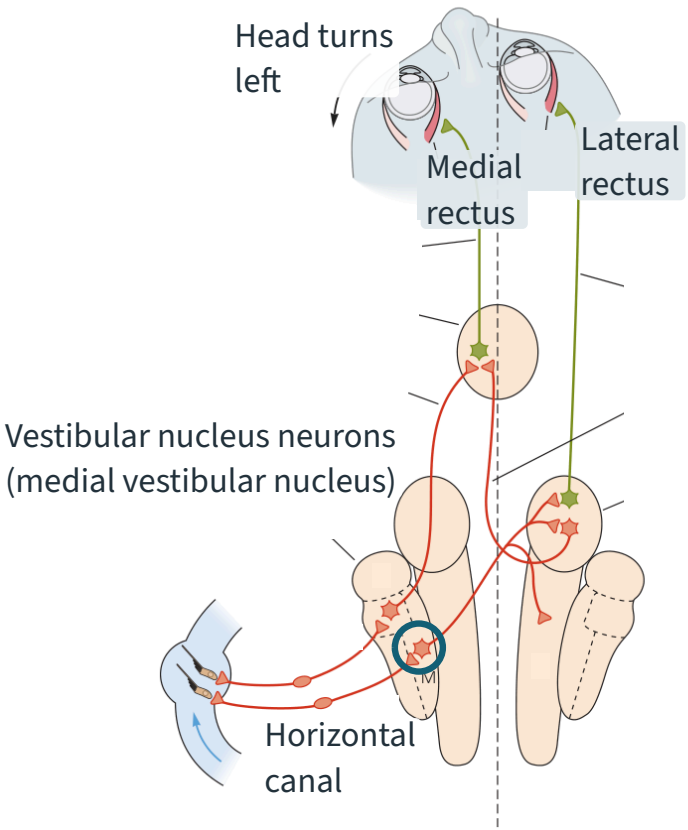
A head turn drives the eyes the opposite way

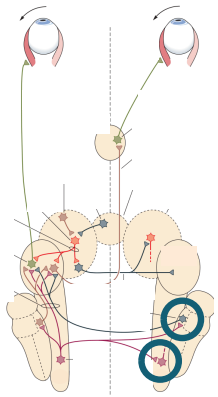


- The canal senses head velocity.

Kandel 5th ed., Fig. 40–11A, p. 927 and p. 926 · excitatory connections of the horizontal reflex, head turning left; Kandel 6e, pp. 869–870 (integrator); labels re-set. Reflex latency: instructor, 2026-11-08

A head turn drives the eyes the opposite way



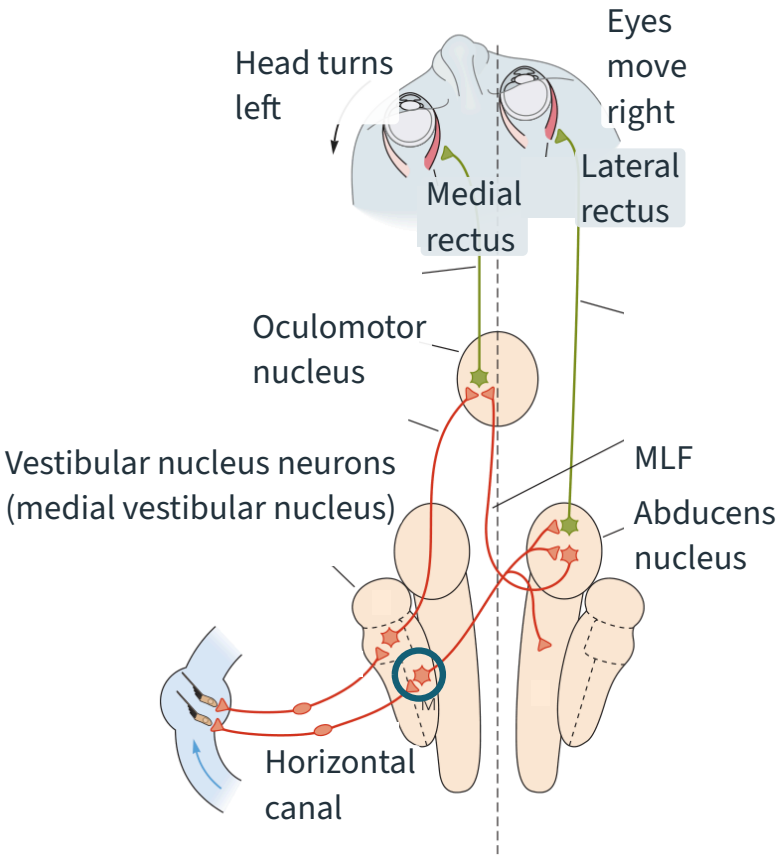


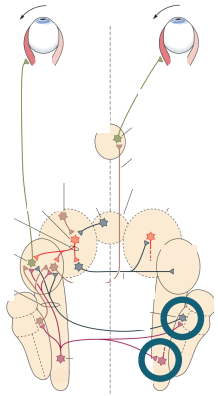
D.06 · integrator: the same nucleus

- The canal senses head velocity.
- Vestibular nucleus neurons relay it: the saccade integrator's nucleus.

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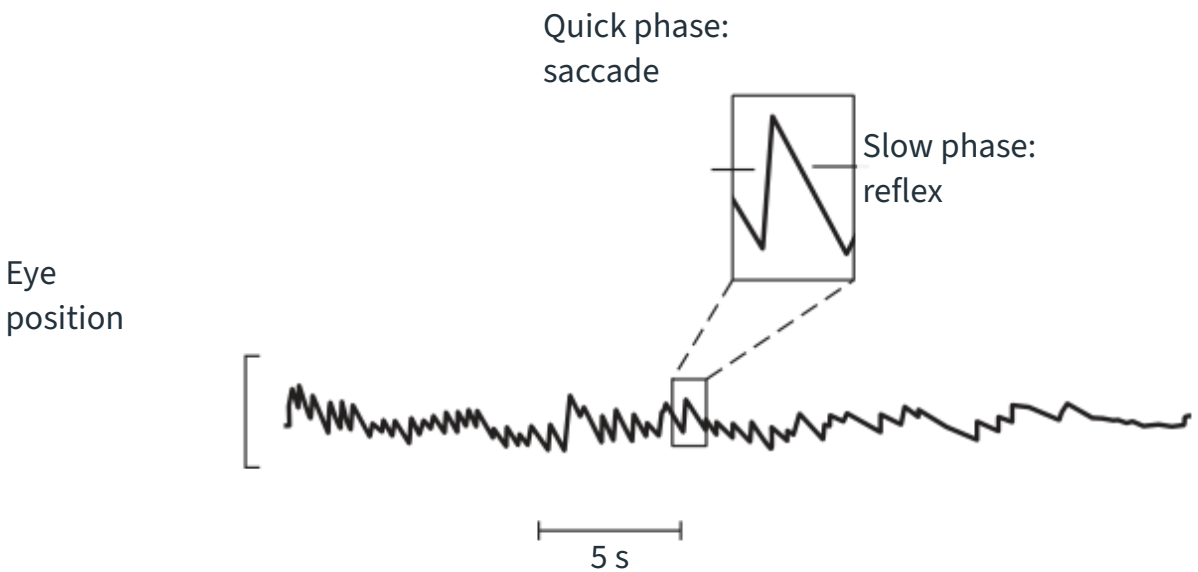


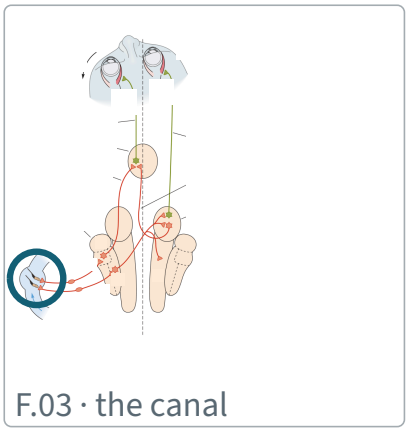
D.06 · integrator: the same nucleus

- The canal senses head velocity.
- Vestibular nucleus neurons relay it: the saccade integrator's nucleus.
- Abducens, and across the MLF oculomotor, neurons turn both eyes against the head.
- Three neurons, 15 ms, no vision: open loop.

Kandel 5th ed., Fig. 40–11A, p. 927 and p. 926 · excitatory connections of the horizontal reflex, head turning left; Kandel 6e, pp. 869–870 (integrator); labels re-set. Reflex latency: instructor, 2026-11-08

Rotation in the dark drives nystagmus



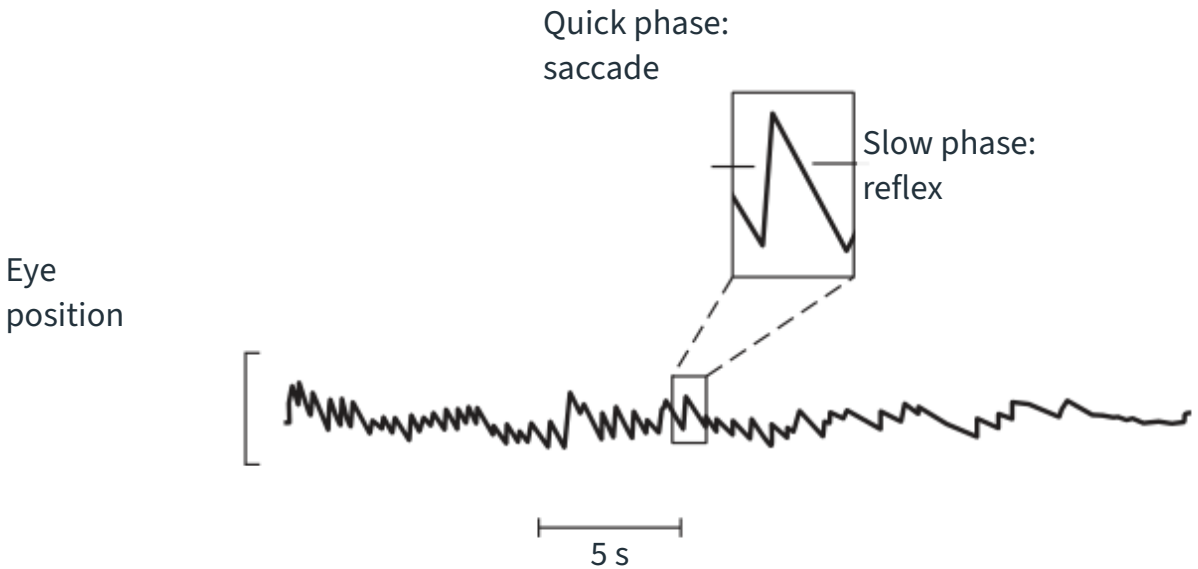


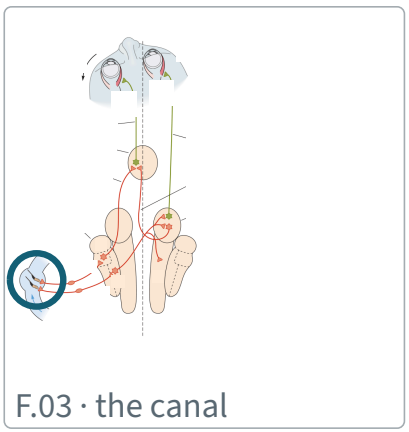
F.03 · the canal

Kandel 5th ed., Fig. 40–8, p. 924 · eye position during constant counterclockwise rotation in the dark (Leigh & Zee 1991); labels re-set

Rotation in the dark drives nystagmus

- The slow phase counter-rotates the eye; a quick phase resets it.

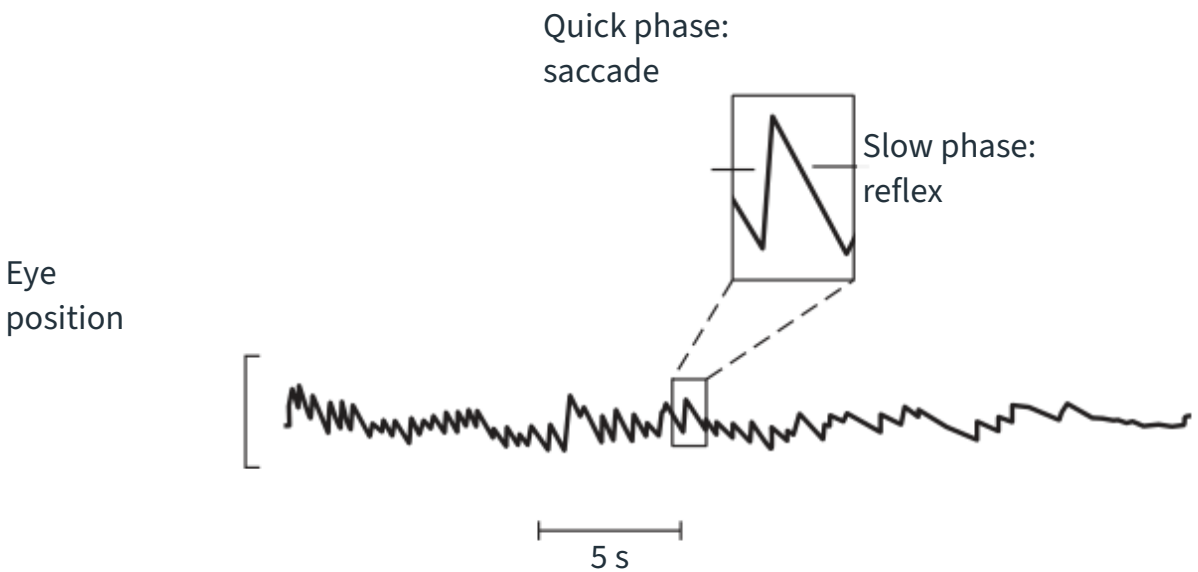




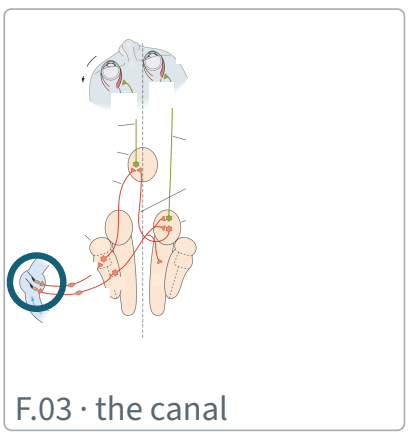
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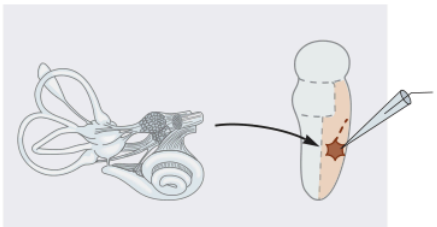
- The slow phase counter-rotates the eye; a quick phase resets it.
- In steady rotation in the dark the canal signal fades, and the nystagmus dies out.



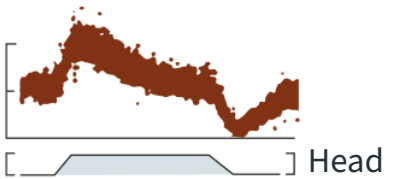
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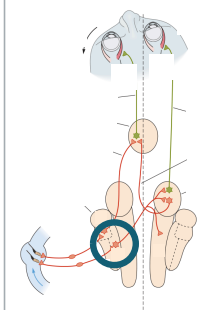
Kandel 5th ed., Fig. 40–8, p. 924 · eye position during constant counterclockwise rotation in the dark (Leigh & Zee 1991); labels re-set

Vision keeps the vestibular nucleus neurons firing



A · Rotation, dark



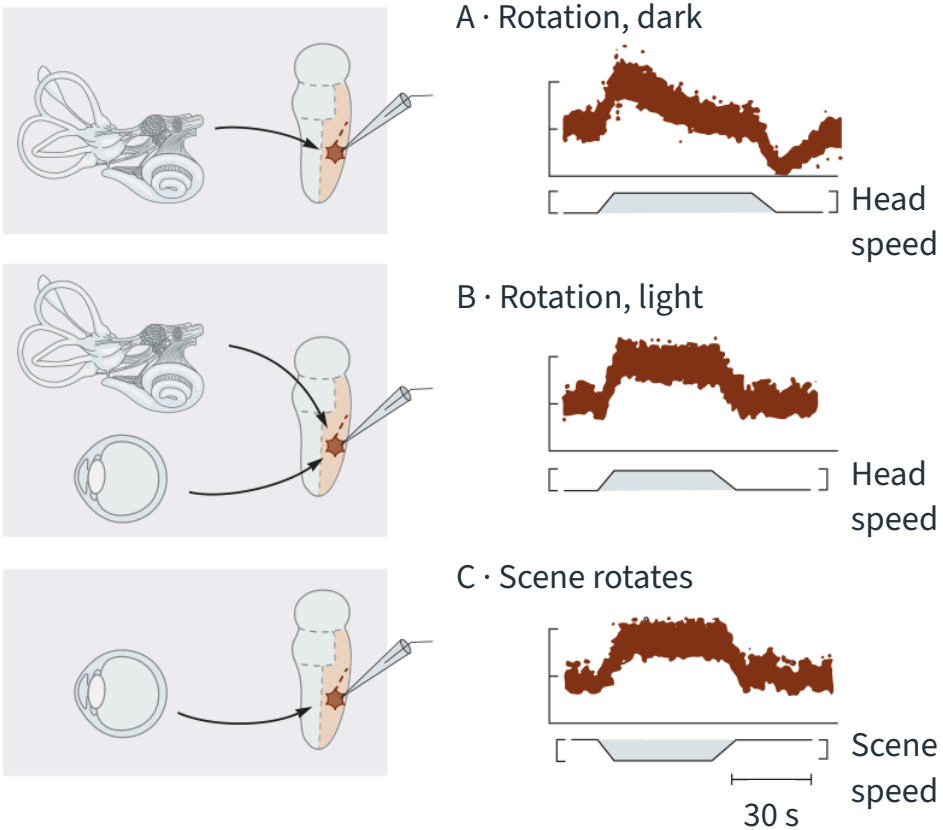


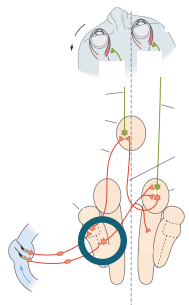
F.03 · vestibular nucleus neurons

- In the dark, firing fades during steady rotation.

Kandel 5th ed., Fig. 40–12, p. 928 and p. 926 · one neuron in the medial vestibular nucleus (Waespe & Henn 1977); firing-rate scales omitted, labels re-set

Vision keeps the vestibular nucleus neurons firing



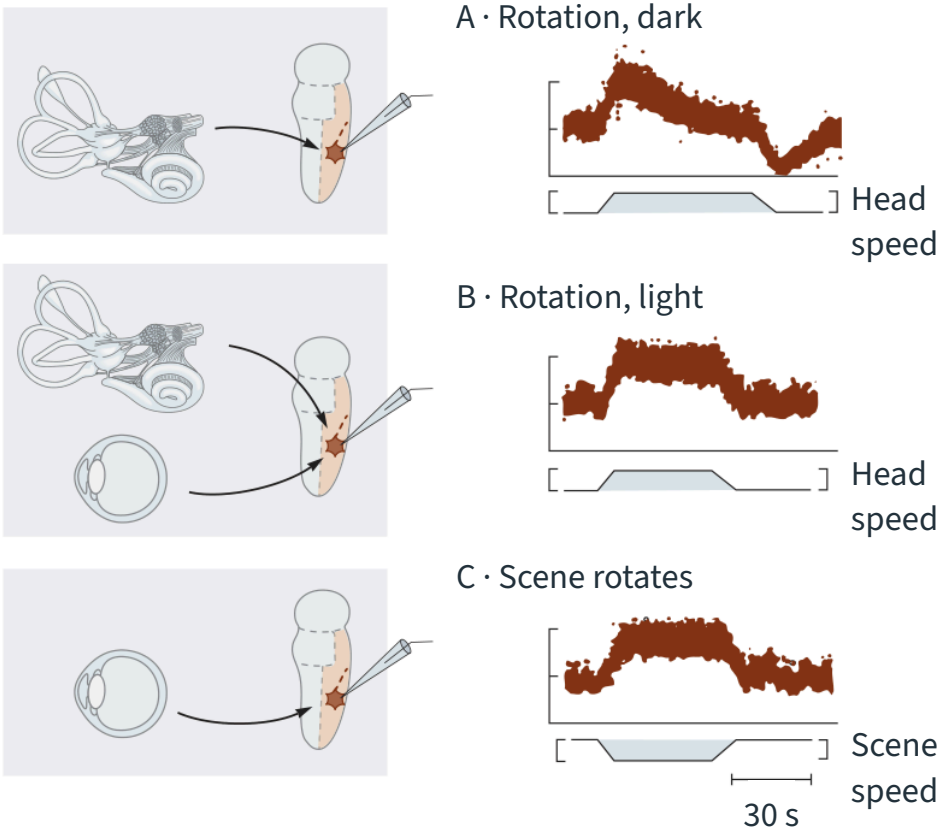


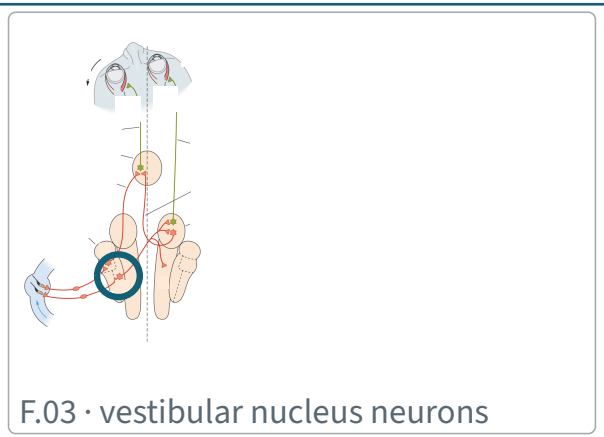
F.03 · vestibular nucleus neurons

- In the dark, firing fades during steady rotation.
- In the light it is sustained, and the scene alone drives it.

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Vision keeps the vestibular nucleus neurons firing



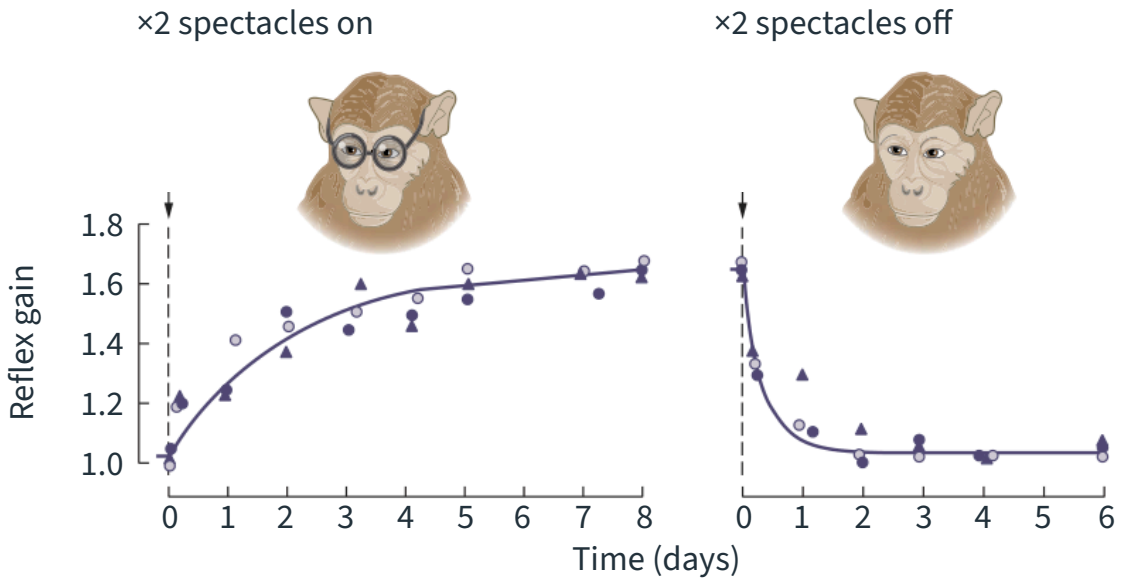


F.03 · vestibular nucleus neurons

- In the dark, firing fades during steady rotation.
- In the light it is sustained, and the scene alone drives it.
- It cannot tell self-motion from scene motion.

Kandel 5th ed., Fig. 40–12, p. 928 and p. 926 · one neuron in the medial vestibular nucleus (Waespe & Henn 1977); firing-rate scales omitted, labels re-set

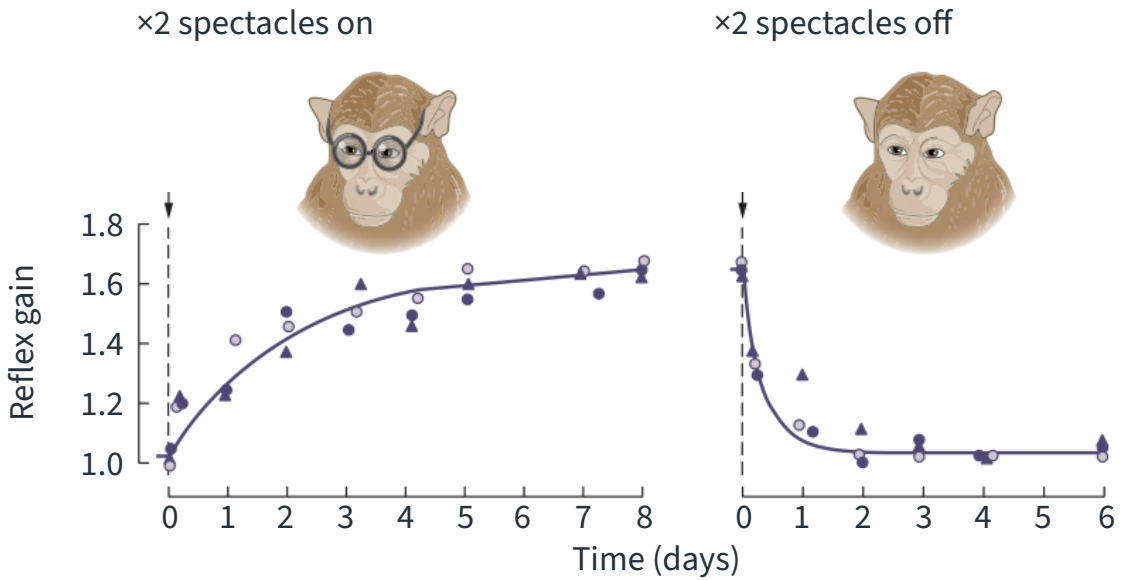
Magnifying spectacles raise the reflex gain



Kandel 5th ed., Fig. 40–14A, p. 930 · reflex gain in a monkey wearing ×2 spectacles, then after removing them (Miles & Eighmy 1980); exposure legend omitted, labels re-set

- Gain climbs toward 2 over days, then falls back.

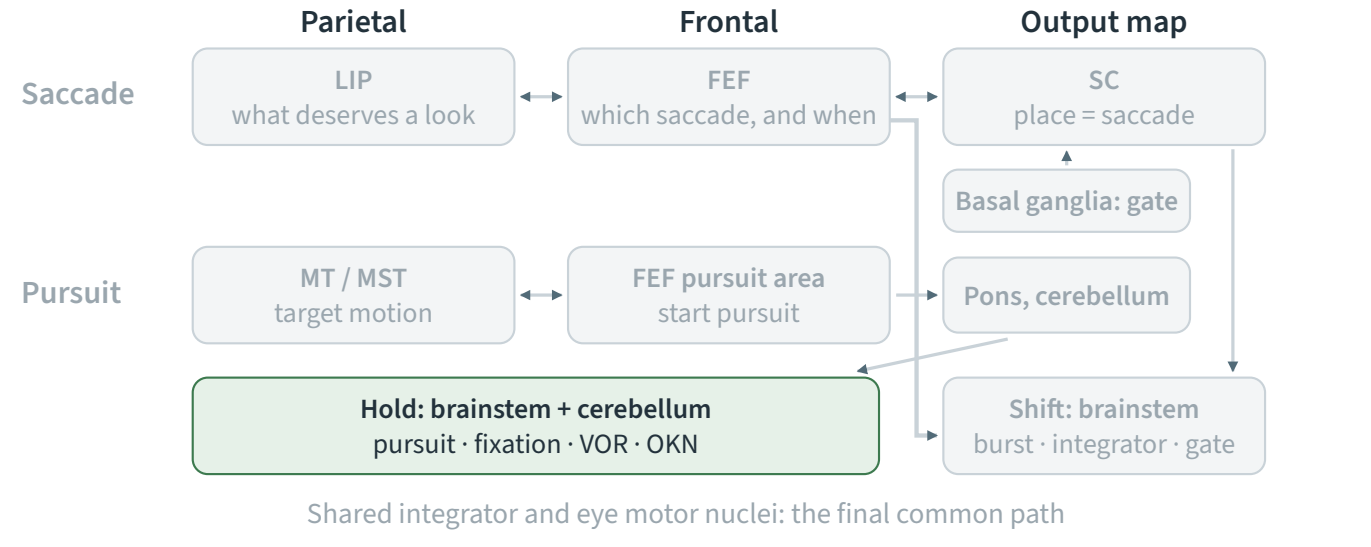
Magnifying spectacles raise the reflex gain



Kandel 5th ed., Fig. 40–14A, p. 930 · reflex gain in a monkey wearing ×2 spectacles, then after removing them (Miles & Eighmy 1980); exposure legend omitted, labels re-set

- Gain climbs toward 2 over days, then falls back.
- The image slip recalibrates the reflex, in the cerebellum.

Section F summary: vision backs up and recalibrates the reflexes

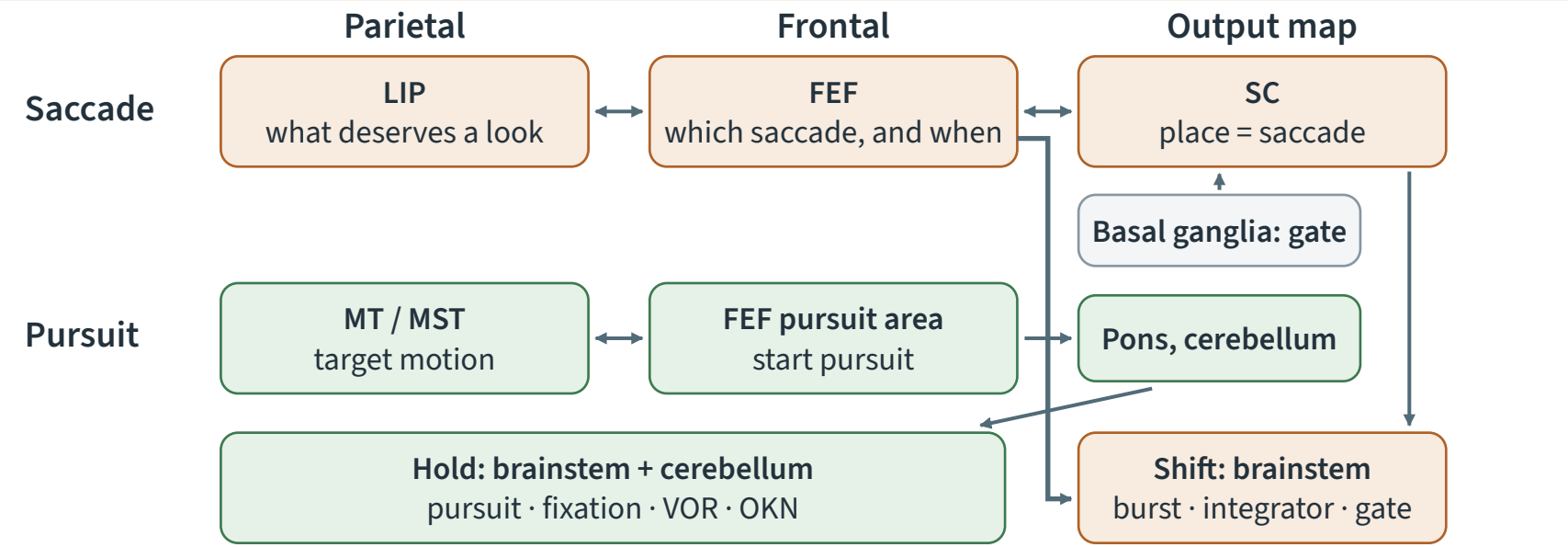


TAKE HOME

- A fast open-loop reflex, backed by slow visual ones, on a shared integrator.
- Vision recalibrates the reflex, through the cerebellum.

Kandel 6e, Ch. 35; Kandel 5th ed., Ch. 40 · the finished map, returning

Lecture summary: a simple effector, controlled like the body



Kandel 6e, Chs. 30, 35, 37; Kandel 5th ed., Chs. 29, 40 · the lecture organizer, complete

TAKE HOME

- The eye is a simple, predictable load: commands need not wait for feedback.
- Gaze is organized like the body: overlapping systems, one final common path.
- Shifting gaze: cortex picks the target, the colliculus turns place into movement, the brainstem builds it.
- Holding gaze: reflexes and pursuit share the integrator, and vision recalibrates them.